Learning the VCF format

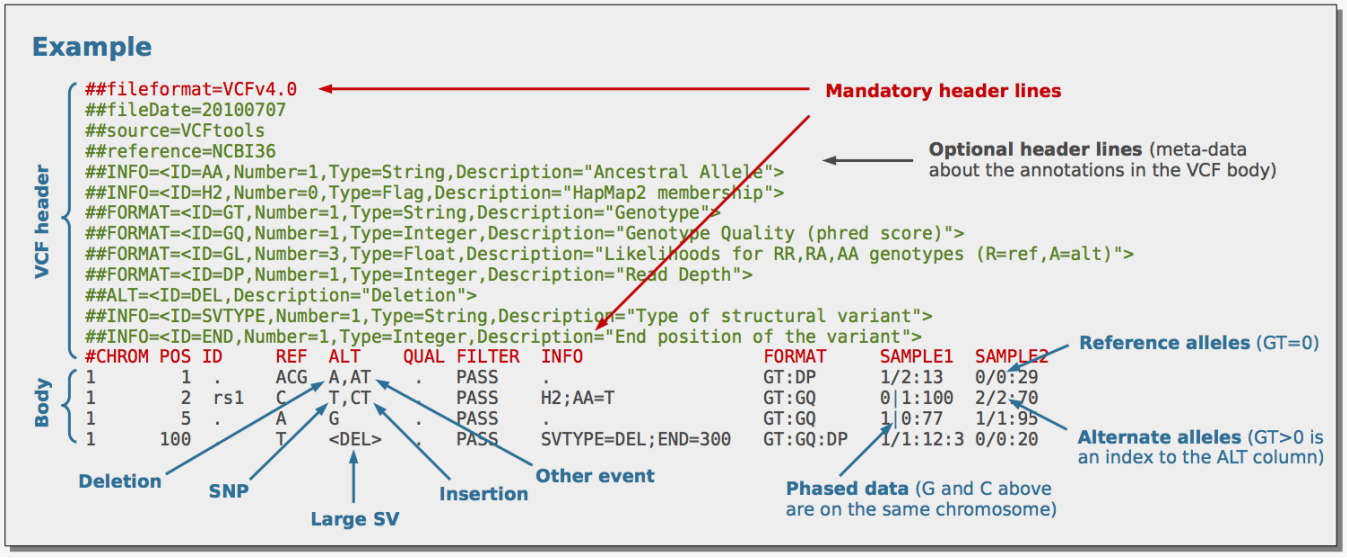
davetang.github.io/learning_vcf_file

Introduction

Natural selection occurs under three conditions:

1. Genetic variation occurs among individuals (and this occurs mainly due to chance errors in replication)
2. The genetic variation must be heritable, i.e. passed from one generation to the next, and
3. The genetic variation results in varying fitness, i.e. individuals survive and reproduce with respect to genetic variation

The *de facto* file format for storing genetic variation is the Variant Call Format (VCF) and was developed under the 1000 Genomes Project. Currently, the Large Scale Genomics work stream of the Global Alliance for Genomics & Health (GA4GH) maintain the specification of the VCF (and other high-throughput sequencing data formats). A good starting point for learning about the VCF is this poster and a portion of the poster is displayed below.



Initially, VCFtools (and the associated scripts) was used for working with VCF files. VCFtools was also developed by the 1000 Genomes Project team but the tool does not seem to be actively maintained anymore. As VCF files are simply tab-delimited flat files, they are slow to process and the BCF was implemented, which is a more efficient format for data processing. A BCF file is the binary equivalent of a VCF file, akin to the SAM and BAM formats and BCFtools is used to work with BCF (and VCF) files. BCFtools is actively maintained and therefore should be used instead of VCFtools. To learn more about BCFtools (and SAMtools), check out the paper Twelve years of SAMtools and BCFtools and please cite it if you use BCFtools for your work.

Lastly, this README is created by `create_readme.sh` using GitHub Actions, which executes `readme.Rmd` and creates `README.md` and can also be viewed at https://davetang.github.io/learning_vcf_file/.

Installation

The easiest way to install BCFtools is by using Conda (and I recommend using Miniconda).

```
conda install -c bioconda bcftools
```

It is also relatively straightforward to compile on Linux (if your system has all the prerequisites). The following code will install BCFtools (and HTSlib and SAMtools, which you will most likely be using as well) into the directory specified by `dir=`.

```
dir=$HOME/tools

ver=1.18
for tool in htslib bcftools samtools; do
   check=${tool}
   if [[ ${tool} == htslib ]]; then
      check=bgzip
   fi
   if [[ ! -e ${dir}/bin/${check} ]]; then

url=https://github.com/samtools/${tool}/releases/download/${ver}/${tool}-${ver}.tar.bz2
      wget ${url}
      tar xjf ${tool}-${ver}.tar.bz2
      cd ${tool}-${ver}
      ./configure --prefix=${dir}
      make && make install
      cd ..
      rm -rf ${tool}-${ver}*
   fi
done
```

Creating VCF example files

Example VCF files were generated to test the functionality of BCFtool and other VCF tools. The `aln.bt.vcf.gz`, `aln.hc.vcf.gz`, and `aln.fb.vcf.gz` VCF files were generated using a simple workflow implemented in Groovy and processed by Bpipe, a tool for running bioinformatics workflows. You can view the workflow in `workflow/simple/pipeline.groovy`, which carries out the following steps: (i) a random reference sequence is generated, (ii) the reference sequence is mutated, (iii) reads are derived from the mutated reference then (iv) mapped back to the original non-mutated reference (v) and finally variants are called using three separate tools: BCFtools, HaplotypeCaller, and freebayes. Additional information is available in this blog post.

To run this workflow for yourself, run the following commands:

```
git clone https://github.com/davetang/learning_vcf_file.git

# installs the necessary tools
cd workflow && ./install.sh

# run the workflow using Bpipe
cd simple && make
```

If you want to play around with different parameters, simply edit the variables at the start of `workflow/simple/pipeline.groovy`. The status of this workflow is checked by [GitHub Actions]() and on successful completion the final variant calls are copied to the `eg` folder; please refer to `.github/workflows/variant_call.yml` for more details.

Usage

Running `bcftools` without any parameters will output the usage and the subcommands. (The `--help` option is used below to avoid an exit code of 1 since this README is generated programmatically.)

```
bcftools --help
```

```
##
## Program: bcftools (Tools for variant calling and manipulating VCFs and BCFs)
## Version: 1.22 (using htslib 1.22)
##
## Usage:   bcftools [--version|--version-only] [--help] <command> <argument>
##
## Commands:
##
##  -- Indexing
##     index        index VCF/BCF files
##
##  -- VCF/BCF manipulation
##     annotate     annotate and edit VCF/BCF files
##     concat       concatenate VCF/BCF files from the same set of samples
##     convert      convert VCF/BCF files to different formats and back
##     head         view VCF/BCF file headers
##     isec         intersections of VCF/BCF files
##     merge        merge VCF/BCF files files from non-overlapping sample sets
##     norm         left-align and normalize indels
##     plugin       user-defined plugins
##     query        transform VCF/BCF into user-defined formats
##     reheader     modify VCF/BCF header, change sample names
##     sort         sort VCF/BCF file
##     view         VCF/BCF conversion, view, subset and filter VCF/BCF files
##
##  -- VCF/BCF analysis
##     call         SNP/indel calling
##     consensus    create consensus sequence by applying VCF variants
##     cnv          HMM CNV calling
##     csq          call variation consequences
##     filter       filter VCF/BCF files using fixed thresholds
##     gtcheck      check sample concordance, detect sample swaps and contamination
##     mpileup      multi-way pileup producing genotype likelihoods
##     roh          identify runs of autozygosity (HMM)
##     stats        produce VCF/BCF stats
##
##  -- Plugins (collection of programs for calling, file manipulation & analysis)
##     41 plugins available, run "bcftools plugin -lv" to see a complete list
##
##  Most commands accept VCF, bgzipped VCF, and BCF with the file type detected
##  automatically even when streaming from a pipe. Indexed VCF and BCF will work
##  in all situations. Un-indexed VCF and BCF and streams will work in most but
##  not all situations.
```

Getting help

If BCFtools was installed by compiling the source code with the path to the man pages added to `MANPATH`, you can simply run `man bcftools` to access the manual pages for getting help. If BCFtools was installed using `Conda`, the `MANPATH` environment variable needs to be set accordingly. The man pages provides additional information for each subcommand and its parameters.

```
man bcftools | grep "LIST OF COMMANDS" -B 1984
```

```
## troff: <standard input>:2458: warning [p 25, 3.3i]: can't break line
## BCFTOOLS(1)                                                    BCFTOOLS(1)
##
## NAME
##        bcftools - utilities for variant calling and manipulating VCFs and
##        BCFs.
##
## SYNOPSIS
##        bcftools [--version|--version-only] [--help] [COMMAND] [OPTIONS]
##
## DESCRIPTION
##        BCFtools  is  a set of utilities that manipulate variant calls in the
##        Variant Call Format (VCF) and its binary counterpart BCF. All commands
##        work transparently with both VCFs and BCFs, both uncompressed and
##        BGZF-compressed.
##
##        Most commands accept VCF, bgzipped VCF and BCF with filetype detected
##        automatically even when streaming from a pipe. Indexed VCF and BCF will
##        work in all situations. Un-indexed VCF and BCF and streams will work in
##        most, but not all situations. In general, whenever multiple VCFs are
##        read simultaneously, they must be indexed and therefore also
##        compressed. (Note that files with non-standard index names can be
##        accessed as e.g. "bcftools view -r X:2928329
##        file.vcf.gz##idx##non-standard-index-name".)
##
##        BCFtools is designed to work on a stream. It regards an input file "-"
##        as the standard input (stdin) and outputs to the standard output
##        (stdout). Several commands can thus be  combined  with  Unix pipes.
##
##    VERSION
##        This manual page was last updated 2025-05-30 and refers to bcftools git
##        version 1.22.
##
##    BCF1
##        The obsolete BCF1 format output by versions of samtools <= 0.1.19 is
##        not compatible with this version of bcftools. To read BCF1 files one
##        can use the view command from old versions of bcftools packaged with
##        samtools versions <= 0.1.19 to convert to VCF, which can then be read
##        by this version of bcftools.
##
##                samtools-0.1.19/bcftools/bcftools view file.bcf1 | bcftools view
##
##    VARIANT CALLING
##        See bcftools call for variant calling from the output of the samtools
##        mpileup command. In versions of samtools <= 0.1.19 calling was done
##        with bcftools view. Users are now required to choose between the old
##        samtools calling model (-c/--consensus-caller) and the new multiallelic
##        calling model (-m/--multiallelic-caller). The multiallelic calling
##        model is recommended for most tasks.
##
##    FILTERING EXPRESSIONS
##        See EXPRESSIONS
##
## LIST OF COMMANDS
```

You can also open an issue and I will try to answer your question.

VCF to BCF and other conversions

Use `bcftools view` or `bcftools convert` to make conversions and use `--output-type u` or `--output-type b` for uncompressed and compressed BCF, respectively.

```
bcftools view -O u eg/aln.bt.vcf.gz > eg/aln.bt.un.bcf
bcftools view -O b eg/aln.bt.vcf.gz > eg/aln.bt.bcf

# uncompressed VCF
bcftools convert -O v eg/aln.bt.vcf.gz > eg/aln.bt.vcf
```

`bcftools convert` has more conversion options such as converting gVCF to VCF and converting VCFs from `IMPUTE2` and `SHAPEIT`.

Viewing a VCF/BCF file

Use `bcftools view` to view a VCF, bgzipped VCF, or BCF file; BCFtools will automatically detect the format.

Viewing a VCF file.

```
bcftools view eg/aln.bt.vcf | grep -v "^#" | head -2

## 1000000  151 .   T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
## 1000000  172 .   TAA TA  142.185 .
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60 GT:PL   1/1:169,117,0
```

Viewing a bgzipped VCF file.

```
bcftools view eg/aln.bt.vcf.gz | grep -v "^#" | head -2

## 1000000  151 .   T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
## 1000000  172 .   TAA TA  142.185 .
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60 GT:PL   1/1:169,117,0
```

Viewing a BCF file.

```
bcftools view eg/aln.bt.vcf.gz | grep -v "^#" | head -2

## 1000000  151 .   T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
## 1000000  172 .   TAA TA  142.185 .
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60 GT:PL   1/1:169,117,0
```

If you want to omit the header but do not want to type `grep -v "^#"`, you can use the `-H` option instead.

```
bcftools view -H eg/aln.bt.vcf.gz | head -2
```

```
## 1000000  151 .   T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
## 1000000  172 .   TAA TA  142.185 .
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60 GT:PL   1/1:169,117,0
```

Comparing output types

The four output types will be compared on a VCF file produced by the 1000 Genomes Project.

1. Compressed BCF (b)
2. Uncompressed BCF (u)
3. Compressed VCF (z)
4. Uncompressed VCF (v).

Download and unzip.

```
wget --quiet
https://davetang.org/file/ALL.chr1.phase3_shapeit2_mvncall_integrated_v5_related_sample
s.20130502.genotypes.vcf.gz -O eg/1kgp.vcf.gz
gunzip eg/1kgp.vcf.gz
```

VCF to compressed BCF.

```
time bcftools convert --threads 2 -O b -o eg/1kgp.bcf eg/1kgp.vcf
```

```
##
## real 0m10.929s
## user 0m24.024s
## sys  0m1.721s
```

VCF to uncompressed BCF.

```
time bcftools convert --threads 2 -O u -o eg/1kgp.un.bcf eg/1kgp.vcf
```

```
##
## real 0m10.509s
## user 0m11.092s
## sys  0m1.475s
```

VCF to compressed VCF.

```
time bcftools convert --threads 2 -O z -o eg/1kgp.vcf.gz eg/1kgp.vcf
```

```
##
## real 0m17.983s
## user 0m29.554s
## sys  0m2.109s
```

File sizes

```
du -h eg/1kgp.*
```

```
## 85M  eg/1kgp.bcf
## 977M eg/1kgp.un.bcf
## 1.5G eg/1kgp.vcf
## 84M  eg/1kgp.vcf.gz
```

VCF to PED

See my blog post.

VCF to BED

VCF to Browser Extensible Data format and not Binary PED format; for Binary PED, see the `plink` directory in this repo. For Browser Extensible Data use BEDOPS, specifically the vcf2bed tool.

Install using Conda from Bioconda.

```
conda install -c bioconda bedops
```

Check out example file.

```
cat eg/ex.vcf | grep -v "^#" | cut -f1-6
```

```
## 1    866511  rs60722469  C   CCCCT   258.62
## 1    884091  rs7522415   C   G   65.46
## 1    897730  rs7549631   C   T   225.34
## 1    1158562 rs57524763  AAC A   220.99
## 1    25563113    .   CC  GG  .
```

Convert to BED.

```
vcf2bed < eg/ex.vcf | cut -f1-3
```

```
## 1    866510  866511
## 1    884090  884091
## 1    897729  897730
## 1    1158561 1158562
## 1    25563112    25563113
```

Note above that the deletion (rs57524763) only has 1 nucleotide but the reference should be 3 nucleotides long. Use `--deletions` to have the coordinates reflect the length of the reference sequence (and to only report deletions).

```
vcf2bed --deletions < eg/ex.vcf | cut -f1-3
```

```
## 1    1158561 1158564
```

There is also the insertions option to report only insertions

```
vcf2bed --insertions < eg/ex.vcf | cut -f1-3
```

```
## 1    866510  866511
```

To report SNVs use `snvs` but note the tool reports the MNP as a SNV and the reference length is not 2 nucleotides long.

```
vcf2bed --snvs < eg/ex.vcf | cut -f1-3
```

```
## 1    884090  884091
## 1    897729  897730
## 1    25563112    25563113
```

Extracting info from columns

Use the `query` subcommand to extract information from the `INFO` or `FORMAT` column.

```
bcftools query -f 'DP=%DP\tAN=%AN\tAC=%AC\tMQ=%MQ\n' eg/aln.bt.vcf.gz | head -3
```

```
## DP=92    AN=2    AC=2    MQ=60
## DP=82    AN=2    AC=2    MQ=60
## DP=109   AN=2    AC=2    MQ=60
```

To extract from the `FORMAT` column, the FORMAT fields must be enclosed in square brackets, e.g. “[%AD]”.

```
bcftools query -f 'AD=[%AD]\tGQ=[%GQ]\tAF=%AF\n' eg/ex.vcf
```

```
## AD=6,56,56,5 GQ=14.7914.7914.79  AF=1
## AD=6,66,66,6 GQ=95.4595.4595.45  AF=0.5
## AD=11,1011,1011,10   GQ=999999   AF=0.5
## AD=14,614,614,6  GQ=999999   AF=0.5
## AD=14,614,614,6  GQ=999999   AF=0.5
```

`query` can read from `STDIN` as with most of the BCFtools commands. The example below extracts information from the `INFO` column for only SNPs.

```
bcftools view -v snps eg/aln.bt.vcf.gz | bcftools query -f
'DP=%DP\tAN=%AN\tAC=%AC\tMQ=%MQ\n' - | head -3
```

```
## DP=92    AN=2    AC=2    MQ=60
## DP=109   AN=2    AC=2    MQ=60
## DP=102   AN=2    AC=2    MQ=60
```

Filtering

Many commands of BCFtools can be used for [filtering variants]() by using the `-f, --apply-filters` or the `-i, --include` and `-e, --exclude` parameters. We will download another example file to demonstrate how to perform filtering.

```
wget --quiet https://davetang.org/file/ERR031940.merged.scored.filtered.bcf -O
eg/ERR031940.bcf
```

To perform filtering on the `FILTER` column of the VCF file, use `bcftools view` and the `-f, --apply-filters` parameter.

```
-f,   --apply-filters <list> require at least one of the listed FILTER strings (e.g.
"PASS,.")
```

Keep variants with `CNN_2D_SNP_Tranche_99.90_100.00` in the `FILTER` column.

```
bcftools view -H -f CNN_2D_SNP_Tranche_99.90_100.00 eg/ERR031940.bcf | wc -l
```

```
## 4505
```

Keep variants with `CNN_2D_SNP_Tranche_99.90_100.00` and `CNN_2D_INDEL_Tranche_99.00_100.00` in the `FILTER` column.

```
bcftools view -H -f CNN_2D_SNP_Tranche_99.90_100.00,CNN_2D_INDEL_Tranche_99.00_100.00
eg/ERR031940.bcf | wc -l
```

```
## 4628
```

Keep variants that have passed filters, i.e. contain `PASS` in the `FILTER` column.

```
bcftools view -H -f PASS eg/ERR031940.bcf | head -3
```

```
## chr1 69270   rs201219564 A   G   689.06  PASS
AC=2;AF=1;AN=2;BaseQRankSum=-2.373;CNN_2D=-5.123;DB;DP=32;ExcessHet=3.0103;FS=0;MLEAC=2
;MLEAF=1;MQ=26.77;MQRankSum=0.472;QD=22.23;ReadPosRankSum=-1.37;SOR=4.615
GT:AD:DP:GQ:PL  1/1:2,29:31:33:703,33,0
## chr1 69511   rs2691305   A   G   1425.06 PASS
AC=2;AF=1;AN=2;BaseQRankSum=1.388;CNN_2D=-2.063;DB;DP=66;ExcessHet=3.0103;FS=4.847;MLEA
C=2;MLEAF=1;MQ=36.21;MQRankSum=0.693;QD=24.57;ReadPosRankSum=0.928;SOR=1.544
GT:AD:DP:GQ:PL  1/1:1,57:58:99:1439,161,0
## chr1 69761   rs200505207 A   T   957.05  PASS
AC=2;AF=1;AN=2;BaseQRankSum=1.919;CNN_2D=-8.226;DB;DP=36;ExcessHet=3.0103;FS=32.516;MLE
AC=2;MLEAF=1;MQ=30.18;MQRankSum=1.342;QD=26.58;ReadPosRankSum=0.086;SOR=5.607
GT:AD:DP:GQ:PL  1/1:3,33:36:26:971,26,0
```

The `filter` command can achieve the same filtering step as above.

```
bcftools filter -i 'FILTER="PASS"' eg/ERR031940.bcf | bcftools view -H - | head -3
```

```
## chr1 69270   rs201219564 A   G   689.06  PASS
AC=2;AF=1;AN=2;BaseQRankSum=-2.373;CNN_2D=-5.123;DB;DP=32;ExcessHet=3.0103;FS=0;MLEAC=2
;MLEAF=1;MQ=26.77;MQRankSum=0.472;QD=22.23;ReadPosRankSum=-1.37;SOR=4.615
GT:AD:DP:GQ:PL  1/1:2,29:31:33:703,33,0
## chr1 69511   rs2691305   A   G   1425.06 PASS
AC=2;AF=1;AN=2;BaseQRankSum=1.388;CNN_2D=-2.063;DB;DP=66;ExcessHet=3.0103;FS=4.847;MLEA
C=2;MLEAF=1;MQ=36.21;MQRankSum=0.693;QD=24.57;ReadPosRankSum=0.928;SOR=1.544
GT:AD:DP:GQ:PL  1/1:1,57:58:99:1439,161,0
## chr1 69761   rs200505207 A   T   957.05  PASS
AC=2;AF=1;AN=2;BaseQRankSum=1.919;CNN_2D=-8.226;DB;DP=36;ExcessHet=3.0103;FS=32.516;MLE
AC=2;MLEAF=1;MQ=30.18;MQRankSum=1.342;QD=26.58;ReadPosRankSum=0.086;SOR=5.607
GT:AD:DP:GQ:PL  1/1:3,33:36:26:971,26,0
```

`bcftools query` can also perform the same filtering using `-i, --include` but a format must be specified.

```
bcftools query -i 'FILTER="PASS"' -f '%CHROM %POS %FILTER\n' eg/ERR031940.bcf | head -3
```

```
## chr1 69270 PASS
## chr1 69511 PASS
## chr1 69761 PASS
```

Which command should be used for filtering? I would recommend using `bcftools filter` for filtering since it clearly states that you are performing filtering in the command name (it is clearer to someone not familiar with BCFtools) and the `-i` and `-e` expressions are flexible enough to perform most filtering requirements. Use `bcftools query` if you want to perform filtering and modify the output. See `man bcftools` for more information about `EXPRESSIONS`, which I recommend reading.

Filtering variant types

Use `TYPE` in the filtering expression to subset specific types of variants; types include: indel, snp, mnp, ref, bnd, other, and overlap.

SNPs.

```
bcftools filter -i 'TYPE="snp"' eg/1kgp.bcf | bcftools view -H - | cut -f1-8 | head -2
```

```
## 1    10505   .   A   T   100 PASS
AC=0;AN=62;NS=31;AF=0.000199681;SAS_AF=0;EUR_AF=0;AFR_AF=0.0008;AMR_AF=0;EAS_AF=0
## 1    10506   .   C   G   100 PASS
AC=0;AN=62;NS=31;AF=0.000199681;SAS_AF=0;EUR_AF=0;AFR_AF=0.0008;AMR_AF=0;EAS_AF=0
```

Indels.

```
bcftools filter -i 'TYPE="indel"' eg/1kgp.bcf | bcftools view -H - | cut -f1-8 | head
-2
```

```
## 1    10177   .   A   AC  100 PASS
AC=19;AN=62;NS=31;AF=0.425319;SAS_AF=0.4949;EUR_AF=0.4056;AFR_AF=0.4909;AMR_AF=0.3602;E
AS_AF=0.3363
## 1    10235   .   T   TA  100 PASS
AC=0;AN=62;NS=31;AF=0.00119808;SAS_AF=0.0051;EUR_AF=0;AFR_AF=0;AMR_AF=0.0014;EAS_AF=0
```

Multiple Nucleotide Polymorphisms: The reference and alternate sequences are of the same length and have to be greater than 1 and all nucleotides in the sequences differ from one another.

```
bcftools filter -i 'TYPE="mnps"' eg/ex.vcf | bcftools view -H - | cut -f1-8
```

```
## 1    25563113    .   CC  GG  .   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=2.621;DB;DP=20;FS=0;HRun=0;HaplotypeScore=101.749;MQ0=0;M
Q=55.8;MQRankSum=-1.91;QD=11.05;ReadPosRankSum=0.4;set=variant
```

Not sure what `ref` refers to but possibly structural variants. `INS:ME` refers to an insertion of a mobile element relative to the reference and the `<CN#>` refers to `Copy number allele: # copies` according to the VCF header.

```
bcftools view -v ref eg/1kgp.bcf | bcftools view -H - | cut -f1-8 | head -3

# filter returns nothing but view does
# bcftools filter -i 'TYPE="ref"' eg/1kgp.bcf | bcftools view -H - | cut -f1-8 | head
-3
```

```
## 1    645710  ALU_umary_ALU_2 A   <INS:ME:ALU>    100 PASS
AC=0;AN=62;CS=ALU_umary;MEINFO=AluYa4_5,1,223,-;NS=31;SVLEN=222;SVTYPE=ALU;TSD=null;AF=
0.00698882;SAS_AF=0.0041;EUR_AF=0.0189;AFR_AF=0;AMR_AF=0.0072;EAS_AF=0.0069
## 1    668630  DUP_delly_DUP20532  G   <CN2>   100 PASS
AC=2;AN=62;CIEND=-150,150;CIPOS=-150,150;CS=DUP_delly;END=850204;NS=31;SVLEN=181574;SVT
YPE=DUP;IMPRECISE;AF=0.0127796;SAS_AF=0.001;EUR_AF=0.001;AFR_AF=0.0015;AMR_AF=0;EAS_AF=
0.0595
## 1    713044  DUP_gs_CNV_1_713044_755966  C   <CN0>,<CN2> 100 PASS
AC=0,2;AN=62;CS=DUP_gs;END=755966;NS=31;SVTYPE=CNV;AF=0.000599042,0.0411342;SAS_AF=0,0.
045;EUR_AF=0.001,0.0417;AFR_AF=0,0.0303;AMR_AF=0.0014,0.0259;EAS_AF=0.001,0.0615
```

Breakends (no variants of this class).

```
bcftools filter -i 'TYPE="bnd"' eg/1kgp.bcf | bcftools view -H -
```

Others.

```
bcftools filter -i 'TYPE="other"' eg/1kgp.bcf | bcftools view -H - | cut -f1-8 | head
-3
```

```
## 1    645710  ALU_umary_ALU_2 A   <INS:ME:ALU>    100 PASS
AC=0;AN=62;CS=ALU_umary;MEINFO=AluYa4_5,1,223,-;NS=31;SVLEN=222;SVTYPE=ALU;TSD=null;AF=
0.00698882;SAS_AF=0.0041;EUR_AF=0.0189;AFR_AF=0;AMR_AF=0.0072;EAS_AF=0.0069
## 1    668630  DUP_delly_DUP20532  G   <CN2>   100 PASS
AC=2;AN=62;CIEND=-150,150;CIPOS=-150,150;CS=DUP_delly;END=850204;NS=31;SVLEN=181574;SVT
YPE=DUP;IMPRECISE;AF=0.0127796;SAS_AF=0.001;EUR_AF=0.001;AFR_AF=0.0015;AMR_AF=0;EAS_AF=
0.0595
## 1    713044  DUP_gs_CNV_1_713044_755966  C   <CN0>,<CN2> 100 PASS
AC=0,2;AN=62;CS=DUP_gs;END=755966;NS=31;SVTYPE=CNV;AF=0.000599042,0.0411342;SAS_AF=0,0.
045;EUR_AF=0.001,0.0417;AFR_AF=0,0.0303;AMR_AF=0.0014,0.0259;EAS_AF=0.001,0.0615
```

Also not sure about `overlap` either.

```
bcftools filter -i 'TYPE="overlap"' eg/1kgp.bcf | bcftools view -H -
```

Filtering genotypes

Use `GT` in the filtering expression to subset specific genotypes.

```
bcftools filter -i 'GT="1|1"' eg/1kgp.bcf | bcftools view -H - | head -3
```

```
## 1    10177   .   A   AC  100 PASS
AC=19;AN=62;NS=31;AF=0.425319;SAS_AF=0.4949;EUR_AF=0.4056;AFR_AF=0.4909;AMR_AF=0.3602;E
AS_AF=0.3363 GT  0|1 0|0 0|0 0|0 1|0 0|0 1|0 0|1 0|0 0|0 0|0 0|0 0|1 0|0 0|1 0|1 1|0
0|1 0|1 0|0 0|0 0|1 0|0 1|0 0|1 1|0 0|0 1|1 1|1 0|0 0|1
## 1    10352   .   T   TA  100 PASS
AC=28;AN=62;NS=31;AF=0.4375;SAS_AF=0.4192;EUR_AF=0.4264;AFR_AF=0.4788;AMR_AF=0.4107;EAS
_AF=0.4306   GT  1|0 0|1 1|1 0|0 1|0 1|0 1|0 1|0 0|0 1|0 1|0 1|0 0|0 0|1 1|0 0|0 1|0
1|1 0|1 0|0 1|0 0|1 1|0 1|0 1|0 0|1 0|1 1|0 0|1 1|0 1|0
## 1    10616   .   CCGCCGTTGCAAAGGCGCGCCG  C   100 PASS
AC=61;AN=62;NS=31;AF=0.993011;SAS_AF=0.9969;EUR_AF=0.994;AFR_AF=0.9894;AMR_AF=0.9957;EA
S_AF=0.9911  GT  1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1
1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|0 1|1 1|1 1|1 1|1 1|1
```

Genotypes using 2nd alternate allele.

```
bcftools filter -i 'GT~"2"' eg/1kgp.bcf | bcftools view -H - | head -3
```

```
## 1    15274   .   A   G,T 100 PASS
AC=25,37;AN=62;NS=31;AF=0.347244,0.640974;SAS_AF=0.3497,0.6472;EUR_AF=0.2922,0.7078;AFR
_AF=0.323,0.6369;AMR_AF=0.2752,0.7205;EAS_AF=0.4812,0.5188   GT  2|1 1|2 2|1 1|2 1|2
2|2 1|1 1|1 2|1 2|2 2|1 2|2 1|2 1|2 2|1 2|2 2|2 2|1 1|2 1|1 2|2 2|1 2|1 2|2 2|2 2|2 1|2
2|1 1|2 2|1 2|1
## 1    565736  .   A   G,T 100 PASS
AC=0,1;AN=62;NS=31;AF=0.000599042,0.00299521;SAS_AF=0,0.0112;EUR_AF=0.001,0;AFR_AF=0,0.
003;AMR_AF=0,0;EAS_AF=0.002,0    GT  0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0
0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 2|0 0|0 0|0 0|0 0|0
## 1    713044  DUP_gs_CNV_1_713044_755966  C   <CN0>,<CN2> 100 PASS
AC=0,2;AN=62;CS=DUP_gs;END=755966;NS=31;SVTYPE=CNV;AF=0.000599042,0.0411342;SAS_AF=0,0.
045;EUR_AF=0.001,0.0417;AFR_AF=0,0.0303;AMR_AF=0.0014,0.0259;EAS_AF=0.001,0.0615 GT
0|0 2|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 2|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0
0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0 0|0
```

You can also use abbreviations such as `GT="AA"`. See below for list of abbreviations.

```
bcftools filter -i 'GT="AA"|GT="RA"' eg/1kgp.bcf | bcftools view -H - | head -3
```

```
## 1    10177   .   A   AC  100 PASS
AC=19;AN=62;NS=31;AF=0.425319;SAS_AF=0.4949;EUR_AF=0.4056;AFR_AF=0.4909;AMR_AF=0.3602;E
AS_AF=0.3363 GT  0|1 0|0 0|0 0|0 1|0 0|0 1|0 0|1 0|0 0|0 0|0 0|0 0|1 0|0 0|1 0|1 1|0
0|1 0|1 0|0 0|0 0|1 0|0 1|0 0|1 1|0 0|0 1|1 1|1 0|0 0|1
## 1    10352   .   T   TA  100 PASS
AC=28;AN=62;NS=31;AF=0.4375;SAS_AF=0.4192;EUR_AF=0.4264;AFR_AF=0.4788;AMR_AF=0.4107;EAS
_AF=0.4306   GT  1|0 0|1 1|1 0|0 1|0 1|0 1|0 1|0 0|0 1|0 1|0 1|0 0|0 0|1 1|0 0|0 1|0
1|1 0|1 0|0 1|0 0|1 1|0 1|0 1|0 0|1 0|1 1|0 0|1 1|0 1|0
## 1    10616   .   CCGCCGTTGCAAAGGCGCGCCG  C   100 PASS
AC=61;AN=62;NS=31;AF=0.993011;SAS_AF=0.9969;EUR_AF=0.994;AFR_AF=0.9894;AMR_AF=0.9957;EA
S_AF=0.9911  GT  1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1
1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|1 1|0 1|1 1|1 1|1 1|1 1|1
```

Below are the list of genotypes you can use:

- `GT="ref"` = reference
- `GT="alt"` = alternate
- `GT="mis"` = missing genotype
- `GT="hom"` = homozygous

- `GT="het"` = heterozygous
- `GT="hap"` = haploid
- `GT="RR"` = ref-ref hom
- `GT="AA"` = alt-alt hom
- `GT="RA"` or GT="AR" = ref-alt het
- `GT="Aa"` or GT="aA" = alt-alt het
- `GT="R"` = haploid ref
- `GT="A"` = haploid alt

Filtering INFO field/s

Use `bcftools filter` to filter out (`-e` or `--exclude`) variants. In the example below we are filtering out variants that have a depth of less than 200. We also use the `-s` parameter to name our filter and this name will be displayed in the `FILTER` column.

```
bcftools filter -s "Depth200" -e "DP<200" eg/aln.bt.vcf.gz | grep -v "^##" | head -4

## #CHROM   POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  test
## 1000000  151 .   T   A   225.417 Depth200
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
## 1000000  172 .   TAA TA  142.185 Depth200
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60 GT:PL   1/1:169,117,0
## 1000000  336 .   A   G   225.417 Depth200
DP=109;VDB=0.972083;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,108,0;MQ=60 GT:PL
1/1:255,255,0
```

Use `&` to combine several other criteria.

```
bcftools filter -s "Depth200&VDB" -e "DP<200 & VDB<0.9" eg/aln.bt.vcf.gz | grep -v
"^##" | head -4

## #CHROM   POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  test
## 1000000  151 .   T   A   225.417 Depth200&VDB
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
## 1000000  172 .   TAA TA  142.185 Depth200&VDB
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60 GT:PL   1/1:169,117,0
## 1000000  336 .   A   G   225.417 PASS
DP=109;VDB=0.972083;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,108,0;MQ=60 GT:PL
1/1:255,255,0
```

Note that `&` and `&&` have different meanings in the expression and make a difference if you have more than one sample in your VCF file. Using `&` requires that the conditions are satisfied within one sample and `&&` means that conditions can be satisfied across samples.

The `-e` or `-i` parameters accept an expression and we can use it to perform calculations. In the example below, variants with a QD divided by DP ratio of less than 0.3 are labelled with `QD/DP`; this ratio was created for illustrative purposes only. Note that `QD` and `DP` are prefixed with `INFO/`;

this was done to explicitly state that we want the QD and DP values from the INFO field, since there is also a DP value in the FORMAT field. (This VCF file is different from the first filtering example, which only had one DP value.)

```
bcftools filter -s "QD/DP" -e "INFO/QD / INFO/DP < 0.3" eg/aln.hc.vcf.gz | grep -v
"^##" | head -4
```

```
## #CHROM   POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  test
## 1000000  151 .   T   A   4111.06 QD/DP
AC=2;AF=1;AN=2;DP=100;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=25.36;SOR=9.065
GT:AD:DP:GQ:PL  1/1:0,92:92:99:4125,277,0
## 1000000  172 .   TA  T   3122.03 PASS
AC=2;AF=1;AN=2;DP=89;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=28.73;SOR=8.909
GT:AD:DP:GQ:PL  1/1:0,85:85:99:3136,256,0
## 1000000  336 .   A   G   4884.06 QD/DP
AC=2;AF=1;AN=2;DP=115;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=30.97;SOR=9.401
GT:AD:DP:GQ:PL  1/1:0,109:109:99:4898,328,0
```

Querying

The `query` and `view` commands can be used to query a VCF file.

Output sample names

Use `bcftools query`.

```
bcftools query -l eg/1kgp.vcf.gz | head -5
```

```
## HG00124
## HG00501
## HG00635
## HG00702
## HG00733
```

Subset sample/s from a multi-sample VCF file

Subset HG00733.

```
bcftools view -s HG00733 eg/1kgp.vcf.gz | grep -v "^##" | head -3
```

```
## #CHROM   POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  HG00733
## 1    10177   .   A   AC  100 PASS
AC=1;AN=2;NS=31;AF=0.425319;SAS_AF=0.4949;EUR_AF=0.4056;AFR_AF=0.4909;AMR_AF=0.3602;EAS
_AF=0.3363   GT  1|0
## 1    10235   .   T   TA  100 PASS
AC=0;AN=2;NS=31;AF=0.00119808;SAS_AF=0.0051;EUR_AF=0;AFR_AF=0;AMR_AF=0.0014;EAS_AF=0
GT  0|0
```

Subset HG00124, HG00501, HG00635, HG00702, and HG00733.

```
bcftools view -s HG00124,HG00501,HG00635,HG00702,HG00733 eg/1kgp.vcf.gz | grep -v "^##"
| head -3
```

```
## #CHROM   POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  HG00124 HG00501 HG00635
HG00702 HG00733
## 1    10177   .   A   AC  100 PASS
AC=2;AN=10;NS=31;AF=0.425319;SAS_AF=0.4949;EUR_AF=0.4056;AFR_AF=0.4909;AMR_AF=0.3602;EA
S_AF=0.3363  GT  0|1 0|0 0|0 0|0 1|0
## 1    10235   .   T   TA  100 PASS
AC=0;AN=10;NS=31;AF=0.00119808;SAS_AF=0.0051;EUR_AF=0;AFR_AF=0;AMR_AF=0.0014;EAS_AF=0
GT  0|0 0|0 0|0 0|0 0|0
```

Subset variants within a specific genomic region

Use `bcftools view` with `-r` or `-R`, which requires an index file. You can use `bcftools view` with `-t` or `-T`, which does not require an index file, but is much slower because the entire file is streamed.

```
tabix -f eg/1kgp.bcf
time bcftools view -H -r 1:55000000-56000000 eg/1kgp.bcf | wc -l
```

```
## 31036
##
## real 0m0.053s
## user 0m0.043s
## sys  0m0.012s
```

`bcftools view` with `-t` streams the entire file, so is much slower.

```
time bcftools view -H -t 1:55000000-56000000 eg/1kgp.bcf | wc -l
```

```
## 31036
##
## real 0m2.841s
## user 0m2.811s
## sys  0m0.032s
```

Use commas to list more than one loci.

```
bcftools view -H -r 1:10000-50000,1:100000-200000,1:55000000-56000000 eg/1kgp.bcf | wc
-l
```

```
## 31382
```

Or use a file (coordinates are 1-based, which is the same as the VCF) to store regions of interest.

```
echo -e "1\t10000\t50000\n1\t100000\t200000\n1\t55000000\t56000000" > eg/regions.txt
bcftools view -H -R eg/regions.txt eg/1kgp.bcf | wc -l
```

```
## 31382
```

Random subset of variants

Total number of variants.

```
bcftools view -H eg/aln.bt.vcf.gz | wc -l
```

```
## 10022
```

A random sample can be achieved by using a Perl one-liner but this is a very slow approach since it goes through every variant line by line. In the example below, the `srand` function sets the seed (for reproducibility) and the float controls how many variants are outputted (1%). (Note that the use of `grep -v "^#"` is only line counting purposes.)

```
bcftools view eg/aln.bt.vcf.gz | perl -nle 'BEGIN { srand(1984) } if (/^#/){ print;
next }; print if rand(1) < 0.01' | grep -v "^#" | wc -l
```

```
## 106
```

Sub-sample 1% and save as BCF file.

```
bcftools view eg/aln.bt.vcf.gz | perl -nle 'BEGIN { srand(1984) } if (/^#/){ print;
next }; print if rand(1) < 0.01' | bcftools view -O b - -o eg/aln.bt.ss.bcf
```

Sample 10%.

```
bcftools view eg/aln.bt.vcf.gz | perl -nle 'BEGIN { srand(1984) } if (/^#/){ print;
next }; print if rand(1) < 0.1' | grep -v "^#" | wc -l
```

```
## 1019
```

Split an annotation field

The split-vep plugin can be used to split a structured field. `split-vep` was written to work with VCF files created by VEP or `bcftools csq`.

```
bcftools +split-vep -h || true
```

```
## split-vep: invalid option -- 'h'
##
## About: Query structured annotations such INFO/BCSQ or CSQ created by bcftools/csq or
VEP. For more
##    more information and pointers see
http://samtools.github.io/bcftools/howtos/plugin.split-vep.html
## Usage: bcftools +split-vep [Plugin Options]
## Plugin options:
##    -a, --annotation STR            INFO annotation to parse, CSQ by default or BCSQ
when not present [CSQ]
##    -A, --all-fields DELIM          Output all fields replacing the -a tag ("%CSQ" by
default) in the -f
##                                      filtering expression using the output field
delimiter DELIM. This can be
##                                      "tab", "space" or an arbitrary string.
##    -c, --columns [LIST|-][:TYPE]   Extract the fields listed either as 0-based
indexes or names, "-" to extract all
##                                      fields. See --columns-types for the defaults.
Supported types are String/Str,
##                                      Integer/Int and Float/Real. Unlisted fields are
set to String. Existing header
##                                      definitions will not be overwritten, remove
first with `bcftools annotate -x`
##        --columns-types -|FILE      Pass "-" to print the default -c types or FILE to
override the presets
##    -d, --duplicate                 Output per transcript/allele consequences on a
new line rather rather than
##                                      as comma-separated fields on a single line
##    -f, --format STR                Create non-VCF output; similar to `bcftools query
-f` but drops lines w/o consequence
##    -g, --gene-list [+]FILE         Consider only features listed in FILE, or
prioritize if FILE is prefixed with "+"
##        --gene-list-fields LIST     Fields to match against by the -g list, by
default gene names [SYMBOL,Gene,gene]
##    -H, --print-header              Print header, -HH to omit column indices
##    -l, --list                      Parse the VCF header and list the annotation
fields
##    -p, --annot-prefix STR          Before doing anything else, prepend STR to all
CSQ fields to avoid tag name conflicts
##    -s, --select TR:CSQ[:PRN]       Select transcripts to extract by type and/or
consequence severity, see also -S and -x
##                                      TR, filter transcripts:
all,worst,primary,pick,mane,EXPRESSION [all]
##                                      CSQ, filter consequences:
any,missense,missense+,etc [any]
##                                      PRN, print consequences:  all,worst [all]
##                                    TR transcript selection
##                                        all       .. list all transcripts
##                                        worst     .. list only one transcript with
the worst consequence (see -S)
##                                        primary   .. list transcripts marked as
CANONICAL=YES
##                                        pick      .. as PICK=1
##                                        mane      .. as MANE_SELECT!=""
##                                      or an EXPRESSION in the form of "<FIELD>
<OPERATOR><VALUE>", where
```

```
##                                         FIELD     .. field name (e.g. "CANONICAL")
##                                         OPERATOR  .. string comparison (=,!=), regex
matching (~,!~)
##                                         VALUE     .. required string value (e.g.
"YES")
##                                     CSQ consequence filtering, selects transcripts by
CSQ severity (see "-S -")
##                                         missense  .. selects only transcripts with a
missense variant
##                                         missense+ .. transcripts with a missense
consequence or more severe
##                                     PRN controls what consequences are actually
printed
##                                         all   .. print all consequences if multiple
per transcript (e.g., start_lost&splice_region)
##                                         worst .. print the worst consequence per
transcript (e.g., start_lost above)
##    -S, --severity -|FILE           Pass "-" to print the default severity scale or
FILE to override
##                                       the default scale
##    -u, --allow-undef-tags          Print "." for undefined tags
##    -x, --drop-sites                Drop sites without consequences after -s is
applied (the default with -f)
##    -X, --keep-sites                Do not drop sites without consequences (the
default without -f)
## Common options:
##    -e, --exclude EXPR              Exclude sites and samples for which the
expression is true
##    -i, --include EXPR              Include sites and samples for which the
expression is true
##        --no-version                Do not append version and command line to the
header
##    -o, --output FILE               Output file name [stdout]
##    -O, --output-type u|b|v|z[0-9]  u/b: un/compressed BCF, v/z: un/compressed VCF,
0-9: compression level [v]
##    -r, --regions REG               Restrict to comma-separated list of regions
##    -R, --regions-file FILE         Restrict to regions listed in a file
##        --regions-overlap 0|1|2     Include if POS in the region (0), record overlaps
(1), variant overlaps (2) [1]
##    -t, --targets REG               Similar to -r but streams rather than index-jumps
##    -T, --targets-file FILE         Similar to -R but streams rather than index-jumps
##        --targets-overlap 0|1|2     Include if POS in the region (0), record overlaps
(1), variant overlaps (2) [0]
##    -v, --verbosity INT             Verbosity level
##    -W, --write-index[=FMT]         Automatically index the output files [off]
##
## Examples:
##    # List available fields of the INFO/CSQ or INFO/BCSQ annotation
##    bcftools +split-vep -l file.vcf.gz
##
##    # List available fields of INFO/BCSQ when both INFO/CSQ and INFO/BCSQ are present
##    bcftools +split-vep -l file.vcf.gz -a BCSQ
##
##    # List the default severity scale
##    bcftools +split-vep -S -
##
```

```
##    # Extract Consequence, IMPACT and gene SYMBOL of the most severe consequence into
##    # INFO annotations starting with the prefix "vep". For brevity, the columns can
##    # be given also as 0-based indexes
##    bcftools +split-vep -c Consequence,IMPACT,SYMBOL -s worst -p vep file.vcf.gz
##    bcftools +split-vep -c 1-3 -s worst -p vep file.vcf.gz
##
##    # Same as above but use the text output of the "bcftools query" format
##    bcftools +split-vep -s worst -f '%CHROM %POS %Consequence %IMPACT %SYMBOL\n'
file.vcf.gz
##
##    # Print all subfields (tab-delimited) in place of %CSQ, each consequence on a new
line
##    bcftools +split-vep -f '%CHROM %POS %CSQ\n' -d -A tab file.vcf.gz
##
##    # Extract gnomAD_AF subfield into a new INFO/gnomAD_AF annotation of Type=Float
so that
##    # numeric filtering can be used.
##    bcftools +split-vep -c gnomAD_AF:Float file.vcf.gz -i'gnomAD_AF<0.001'
##
##    # Similar to above, but add the annotation only if the consequence severity is
missense
##    # or equivalent. In order to drop sites with different consequences completely,
provide
##    # the -x option. See the online documentation referenced above for more examples.
##    bcftools +split-vep -c gnomAD_AF:Float -s :missense    file.vcf.gz
##    bcftools +split-vep -c gnomAD_AF:Float -s :missense -x file.vcf.gz
##
##    See also http://samtools.github.io/bcftools/howtos/plugin.split-vep.html
```

VEP

An [example VCF file]() that was annotated with VEP is available as `eg/S1.haplotypecaller.filtered_VEP.ann.vcf.gz`. To list the annotation fields use `-l`.

```
bcftools +split-vep -l eg/S1.haplotypecaller.filtered_VEP.ann.vcf.gz | head
```

```
## 0    Allele
## 1    Consequence
## 2    IMPACT
## 3    SYMBOL
## 4    Gene
## 5    Feature_type
## 6    Feature
## 7    BIOTYPE
## 8    EXON
## 9    INTRON
```

Use `-f` to print the wanted fields in your own specified format; variants without consequences are excluded.

```
bcftools +split-vep -f '%CHROM:%POS,%ID,%Consequence\n'
eg/S1.haplotypecaller.filtered_VEP.ann.vcf.gz | head
```

```
## chr1:69270,rs201219564,regulatory_region_variant,synonymous_variant
## chr1:69511,rs2691305,missense_variant
## chr1:69761,rs200505207,missense_variant
## chr1:69897,rs200676709,synonymous_variant
##
chr1:941119,rs4372192,regulatory_region_variant,intron_variant,downstream_gene_variant
## chr1:946247,rs2272757,downstream_gene_variant,synonymous_variant
## chr1:952180,rs3748595,regulatory_region_variant,intron_variant
## chr1:952421,rs3828047,synonymous_variant
## chr1:953259,rs3748596,synonymous_variant
## chr1:953279,rs3748597,missense_variant
```

Limit output to missense or more severe variants.

```
bcftools +split-vep -f '%CHROM:%POS,%ID,%Consequence\n' -s worst:missense+
eg/S1.haplotypecaller.filtered_VEP.ann.vcf.gz | head
```

```
## chr1:69511,rs2691305,missense_variant
## chr1:69761,rs200505207,missense_variant
## chr1:953279,rs3748597,missense_variant
## chr1:953778,rs13303056,splice_donor_region_variant&intron_variant
## chr1:953779,rs13302945,splice_donor_region_variant&intron_variant
## chr1:973858,rs3829740,missense_variant
## chr1:973929,rs3829738,missense_variant
##
chr1:1041950,rs2799066,splice_polypyrimidine_tract_variant&splice_region_variant&intron
_variant
## chr1:1047561,rs3128102,splice_polypyrimidine_tract_variant&intron_variant
## chr1:1051820,rs9803031,splice_donor_5th_base_variant&intron_variant
```

BCFtools csq

An [example VCF file] that was annotated with BCFtools csq is available as `eg/S1.haplotypecaller.filtered.phased.csq.vcf.gz`. The tag added by `csq` is `INFO/BCSQ`, so we need to provide this to split-vep. To list the annotation fields use `-l`.

```
bcftools +split-vep -a BCSQ -l eg/S1.haplotypecaller.filtered.phased.csq.vcf.gz
```

```
## 0    Consequence
## 1    gene
## 2    transcript
## 3    biotype
## 4    strand
## 5    amino_acid_change
## 6    dna_change
```

Use `-f` to print the wanted fields in your own specified format; variants without consequences are excluded.

```
bcftools +split-vep -a BCSQ -f '%CHROM:%POS,%ID,%Consequence\n'
eg/S1.haplotypecaller.filtered.phased.csq.vcf.gz | head
```

```
## chr1:69270,rs201219564,synonymous
## chr1:69511,rs2691305,missense
## chr1:69761,rs200505207,missense
## chr1:69897,rs200676709,synonymous
## chr1:941119,rs4372192,intron,non_coding
## chr1:946247,rs2272757,synonymous
## chr1:952180,rs3748595,intron,non_coding
## chr1:952421,rs3828047,synonymous
## chr1:953259,rs3748596,synonymous
## chr1:953279,rs3748597,missense
```

The `-d` or `--duplicate` is useful to output annotations per transcript/allele on a new line.

```
bcftools +split-vep -a BCSQ -f '%transcript,%Consequence\n'
eg/S1.haplotypecaller.filtered.phased.csq.vcf.gz | head
```

```
## Warning: fewer BCSQ fields than expected at chr1:13226098, filling with dots. This
warning is printed only once.
## ENST00000641515,synonymous
## ENST00000641515,missense
## ENST00000641515,missense
## ENST00000641515,synonymous
## .,.,intron,non_coding
## ENST00000327044,synonymous
## .,.,intron,non_coding
## ENST00000327044,synonymous
## ENST00000327044,synonymous
## ENST00000327044,missense
```

Use `-d` to split.

```
bcftools +split-vep -a BCSQ -d -f '%transcript,%Consequence\n'
eg/S1.haplotypecaller.filtered.phased.csq.vcf.gz | head
```

```
## Warning: fewer BCSQ fields than expected at chr1:13226098, filling with dots. This
warning is printed only once.
## ENST00000641515,synonymous
## ENST00000641515,missense
## ENST00000641515,missense
## ENST00000641515,synonymous
## .,intron
## .,non_coding
## ENST00000327044,synonymous
## .,intron
## .,non_coding
## ENST00000327044,synonymous
```

SnpEff

It is possible to use the split-vep plugin with SnpEff, another popular variant annotation tool with some modifications to the VCF header.

SnpEff provides annotations with the `ANN` tag.

```
bcftools view -H eg/PRJNA784038_illumina.vt.ann.vcf.gz | head -1 | cut -f8
```

```
##
VDB=0.02;SGB=-0.453602;RPBZ=1.63951;MQBZ=0;MQSBZ=0;BQBZ=-1.38042;SCBZ=-0.632456;FS=0;MQ
0F=0;MQ=60;DP=8;DP4=2,3,1,1;AN=2;AC=1;ANN=T|upstream_gene_variant|MODIFIER|ORF1ab|GU280
_gp01|transcript|GU280_gp01|protein_coding||c.-262A>T|||||262|,T|upstream_gene_variant|
MODIFIER|ORF1ab|GU280_gp01|transcript|YP_009725297.1|protein_coding||c.-262A>T|||||262|
WARNING_TRANSCRIPT_NO_STOP_CODON,T|upstream_gene_variant|MODIFIER|ORF1ab|GU280_gp01|tra
nscript|YP_009742608.1|protein_coding||c.-262A>T|||||262|WARNING_TRANSCRIPT_NO_STOP_COD
ON,T|upstream_gene_variant|MODIFIER|ORF1ab|GU280_gp01|transcript|GU280_gp01.2|protein_c
oding||c.-262A>T|||||262|,T|upstream_gene_variant|MODIFIER|ORF1ab|GU280_gp01|transcript
|YP_009725298.1|protein_coding||c.-802A>T|||||802|WARNING_TRANSCRIPT_NO_START_CODON,T|u
pstream_gene_variant|MODIFIER|ORF1ab|GU280_gp01|transcript|YP_009742609.1|protein_codin
g||c.-802A>T|||||802|WARNING_TRANSCRIPT_NO_START_CODON,T|upstream_gene_variant|MODIFIER
|ORF1ab|GU280_gp01|transcript|YP_009725299.1|protein_coding||c.-2716A>T|||||2716|WARNIN
G_TRANSCRIPT_NO_START_CODON,T|upstream_gene_variant|MODIFIER|ORF1ab|GU280_gp01|transcri
pt|YP_009742610.1|protein_coding||c.-2716A>T|||||2716|WARNING_TRANSCRIPT_NO_START_CODON
,T|intergenic_region|MODIFIER|CHR_START-ORF1ab|CHR_START-
GU280_gp01|intergenic_region|CHR_START-GU280_gp01|||n.4A>T||||||
```

We can tell `split-vep` to use the `ANN` field using the `-a` parameter but this will result in an error because the header is not defined in the expected format. (The `|| true` is added to prevent GitHub Actions from failing and you do not normally need it.)

```
bcftools +split-vep eg/PRJNA784038_illumina.vt.ann.vcf.gz -a ANN || true

## Expected "Format: " substring in the header INFO/ANN/Description, found: "Functional
annotations: 'Allele | Annotation | Annotation_Impact | Gene_Name | Gene_ID |
Feature_Type | Feature_ID | Transcript_BioType | Rank | HGVS.c | HGVS.p | cDNA.pos /
cDNA.length | CDS.pos / CDS.length | AA.pos / AA.length | Distance | ERRORS / WARNINGS
/ INFO'"
```

The [expected format](#) is:

```
##INFO=<ID=CSQ,Number=.,Type=String,Description="Consequence annotations from Ensembl
VEP. Format:
Allele|Consequence|IMPACT|SYMBOL|Gene|Feature_type|Feature|BIOTYPE|EXON|INTRON|HGVSc|HG
VSp|cDNA_position|CDS_position|Protein_position">
#CHROM  POS       ID          REF  ALT  QUAL  FILTER  INFO
21      26978790  rs75377686  T    C    .     .
CSQ=C|missense_variant|MODERATE|MRPL39|ENSG00000154719|Transcript|ENST00000419219|prote
in_coding|2/8||ENST00000419219.1:c.251A>G|ENSP00000404426.1:p.Asn84Ser|260|251|84
```

We can use `bcftools reheader` to modify the header for `ANN` so that is `split-vep` compatible.

```
bcftools head eg/PRJNA784038_illumina.vt.ann.vcf.gz | perl -nle "if (/ID=ANN,/){
s/[\s\']//g; s/Functionalannotations:/Format: /; s/Annotation_Impact/Consequence/ }
print" > eg/new_header.txt
bcftools reheader -h eg/new_header.txt eg/PRJNA784038_illumina.vt.ann.vcf.gz -o
eg/PRJNA784038_illumina.vt.ann.vep.bcf
```

We can now run `split-vep` and list the split annotation fields.

```
bcftools +split-vep -a ANN -l eg/PRJNA784038_illumina.vt.ann.vep.bcf
```

```
## 0    Allele
## 1    Annotation
## 2    Consequence
## 3    Gene_Name
## 4    Gene_ID
## 5    Feature_Type
## 6    Feature_ID
## 7    Transcript_BioType
## 8    Rank
## 9    HGVS.c
## 10   HGVS.p
## 11   cDNA.pos_cDNA.length
## 12   CDS.pos_CDS.length
## 13   AA.pos_AA.length
## 14   Distance
## 15   ERRORS_WARNINGS_INFO
```

This variant file contains Omicron variants called using [BCFtools](). We can use `split-vep` and `awk` to output Spike protein variants that have a `LOW` or higher consequence. The `-d` parameter is used to output consequences on a new line rather than having comma-delimited entries.

```
bcftools +split-vep -a ANN -d -f
'%CHROM:%POS\t%Gene_Name\t%Consequence\t%HGVS.c\t%HGVS.p\n'
eg/PRJNA784038_illumina.vt.ann.vep.bcf | awk '$2 == "S" && $3 != "MODIFIER" {print}' |
head
```

```
## NC_045512.2:21588    S   MODERATE    c.26C>T p.Pro9Leu
## NC_045512.2:21636    S   MODERATE    c.74C>T p.Pro25Leu
## NC_045512.2:21736    S   LOW c.174C>T    p.Phe58Phe
## NC_045512.2:21754    S   MODERATE    c.192G>T    p.Trp64Cys
## NC_045512.2:21762    S   MODERATE    c.200C>T    p.Ala67Val
## NC_045512.2:21768    S   MODERATE    c.206A>T    p.His69Leu
## NC_045512.2:21770    S   MODERATE    c.208G>A    p.Val70Ile
## NC_045512.2:21770    S   MODERATE    c.208G>C    p.Val70Leu
## NC_045512.2:21770    S   MODERATE    c.208G>T    p.Val70Phe
## NC_045512.2:21846    S   MODERATE    c.284C>T    p.Thr95Ile
```

Summarise SNPs and INDELs per sample

Use `bcftools stats` with the `-s -` parameter. The example VCF file `eg/ex.vcf` has four variants across three samples (one, two, and three).

- Sample one has two SNPs (both het) and one deletion (het)
- Sample two has two SNPs (one het and one hom alt) and insertion (het)
- Sample three has one SNP (het) and one insertion (hom alt) and one deletion (hom alt)

You can confirm the numbers from the stats output.

```
cat eg/ex.vcf | grep -v "^#"
```

```
## 1    866511  rs60722469  C   CCCCT   258.62  PASS
AC=2;AF=1.00;AN=2;DB;DP=11;FS=0.000;HRun=0;HaplotypeScore=41.3338;MQ0=0;MQ=61.94;QD=23.
51;set=variant   GT:AD:DP:GQ:PL  0/0:6,5:11:14.79:300,15,0   0/1:6,5:11:14.79:300,15,0
1/1:6,5:11:14.79:300,15,0
## 1    884091  rs7522415   C   G   65.46   PASS
AC=1;AF=0.50;AN=2;BaseQRankSum=-0.259;DB;DP=12;Dels=0.00;FS=0.000;HRun=1;HaplotypeScore
=0.0000;MQ0=0;MQ=53.22;MQRankSum=0.779;QD=5.45;ReadPosRankSum=2.047;set=variant2
GT:AD:DP:GQ:PL  0/1:6,6:12:95.45:95,0,123   1/1:6,6:12:95.45:95,0,123
0/0:6,6:12:95.45:95,0,123
## 1    897730  rs7549631   C   T   225.34  PASS
AC=1;AF=0.50;AN=2;BaseQRankSum=-2.218;DB;DP=21;Dels=0.00;FS=6.419;HRun=1;HaplotypeScore
=1.8410;MQ0=0;MQ=58.89;MQRankSum=-0.387;QD=10.73;ReadPosRankSum=-0.880;set=variant2
GT:AD:DP:GQ:PL  0/1:11,10:21:99:255,0,348   0/1:11,10:21:99:255,0,348
0/1:11,10:21:99:255,0,348
## 1    1158562 rs57524763  AAC A   220.99  PASS
AC=1;AF=0.50;AN=2;BaseQRankSum=2.621;DB;DP=20;FS=0.000;HRun=0;HaplotypeScore=101.7487;M
Q0=0;MQ=55.80;MQRankSum=-1.910;QD=11.05;ReadPosRankSum=0.400;set=variant GT:AD:DP:GQ:PL
0/1:14,6:20:99:260,0,486    0/0:14,6:20:99:260,0,486    1/1:14,6:20:99:260,0,486
## 1    25563113    .   CC  GG  .   PASS
AC=1;AF=0.50;AN=2;BaseQRankSum=2.621;DB;DP=20;FS=0.000;HRun=0;HaplotypeScore=101.7487;M
Q0=0;MQ=55.80;MQRankSum=-1.910;QD=11.05;ReadPosRankSum=0.400;set=variant GT:AD:DP:GQ:PL
0/1:14,6:20:99:260,0,486    0/0:14,6:20:99:260,0,486    1/1:14,6:20:99:260,0,486
```

```
bcftools stats -s - eg/ex.vcf | grep -A 4 "Per-sample counts"
```

```
## # PSC, Per-sample counts. Note that the ref/het/hom counts include only SNPs, for
indels see PSI. The rest include both SNPs and indels.
## # PSC    [2]id   [3]sample   [4]nRefHom  [5]nNonRefHom   [6]nHets    [7]nTransitions
[8]nTransversions   [9]nIndels  [10]average depth   [11]nSingletons [12]nHapRef
[13]nHapAlt [14]nMissing
## PSC  0   one 1   0   2   1   1   1   16.8    0   0   0   0
## PSC  0   two 2   1   1   1   1   1   16.8    0   0   0   0
## PSC  0   three   1   0   1   1   0   2   16.8    0   0   0   0
```

```
bcftools stats -s - eg/ex.vcf | grep -A 4 "Per-Sample Indels"
```

```
## # PSI, Per-Sample Indels. Note that alt-het genotypes with both ins and del allele
are counted twice, in both nInsHets and nDelHets.
## # PSI    [2]id   [3]sample   [4]in-frame [5]out-frame    [6]not applicable
[7]out/(in+out) ratio   [8]nInsHets [9]nDelHets [10]nInsAltHoms [11]nDelAltHoms
## PSI  0   one 0   0   0   0.00    0   1   0   0
## PSI  0   two 0   0   0   0.00    1   0   0   0
## PSI  0   three   0   0   0   0.00    0   0   1   1
```

Add AF tag to a VCF file

The allele frequency tag (AF) is missing from the VCF file generated by `BCFtools call`.

```
bcftools view -H eg/aln.bt.vcf.gz | head -1
```

```
## 1000000  151 .   T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
```

The `fill-tags` plugin can add additional tags to a VCF file including the AF tag.

```
bcftools plugin fill-tags eg/aln.bt.vcf.gz | grep -v "^#" | head -1
```

```
## 1000000  151 .   T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60;F_MISSING=0;N
S=1;AF=1;MAF=0;AC_Het=0;AC_Hom=2;AC_Hemi=0;HWE=1;ExcHet=1    GT:PL   1/1:255,255,0
```

Add custom annotations

The `annotate` function can be used to [add additional annotations]() to a VCF file. You can use a VCF file or a tabix-indexed tab-delimited file; if a tab-delimited file is used, an additional header file is required.

To demonstrate, a tab-delimited file with incremental integers (fifth column) is created.

```
bcftools view -H eg/aln.bt.vcf.gz | head -3 | cut -f1,2,4,5 | perl -nle '$i++; print
join("\t", $_, "$i")' > eg/test_anno.tsv
cat eg/test_anno.tsv
```

```
## 1000000  151 T   A   1
## 1000000  172 TAA TA  2
## 1000000  336 A   G   3
```

`eg/test_anno.tsv` needs to be tabix-indexed with the following parameters:

- `-s` specifies the column for sequence names
- `-b` specifies the column for region start/s
- `-e` specifies the column for region end/s

```
bgzip eg/test_anno.tsv
tabix -s1 -b2 -e2 eg/test_anno.tsv.gz
```

Create header file.

```
echo -e "##INFO=<ID=INC,Number=1,Type=Integer,Description=\"Increment number\">" >
eg/test_anno.hdr
```

Add `INC` annotation.

```
bcftools annotate -a eg/test_anno.tsv.gz -h eg/test_anno.hdr -c CHROM,POS,REF,ALT,INC
eg/aln.bt.vcf.gz | grep -v "^##" | head -4
```

```
## #CHROM   POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  test
## 1000000  151 .   T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60;INC=1 GT:PL
1/1:255,255,0
## 1000000  172 .   TAA TA  142.185 .
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60;INC=2   GT:PL   1/1:169,117,0
## 1000000  336 .   A   G   225.417 .
DP=109;VDB=0.972083;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,108,0;MQ=60;INC=3
GT:PL   1/1:255,255,0
```

To add the custom annotations to the ID column of the VCF file, specify the ID column instead of the INFO tag with the `-c` parameter. (Note that we no longer require the header file, since the ID column is a standard column in the VCF.)

```
bcftools annotate -a eg/test_anno.tsv.gz -c CHROM,POS,REF,ALT,ID eg/aln.bt.vcf.gz |
grep -v "^##" | head -4
```

```
## #CHROM   POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  test
## 1000000  151 1   T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
## 1000000  172 2   TAA TA  142.185 .
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60 GT:PL   1/1:169,117,0
## 1000000  336 3   A   G   225.417 .
DP=109;VDB=0.972083;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,108,0;MQ=60 GT:PL
1/1:255,255,0
```

In addition, `annotate` can set IDs on-the-fly using information from the VCF columns using the `--set-id` parameter.

```
bcftools annotate --set-id '%CHROM\_%POS\_%REF\_%FIRST_ALT' eg/aln.bt.vcf.gz | grep -v
"^##" | head -4
```

```
## #CHROM   POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  test
## 1000000  151 1000000_151_T_A T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
## 1000000  172 1000000_172_TAA_TA  TAA TA  142.185 .
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60 GT:PL   1/1:169,117,0
## 1000000  336 1000000_336_A_G A   G   225.417 .
DP=109;VDB=0.972083;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,108,0;MQ=60 GT:PL
1/1:255,255,0
```

Check whether the REF sequence is correct

Change the reference sequence of variant at position 151 to G.

```
zcat eg/aln.bt.vcf.gz | perl -lane 'if($F[1] == 151){ $F[3] = G; print join("\t", @F) }
else { print }' > eg/incorrect.vcf
```

Use `bcftools norm` with `-c` to check whether the REF sequence is correct.

```
bcftools norm -f eg/test_31.fa -c w eg/incorrect.vcf > /dev/null
```

```
## REF_MISMATCH 1000000 151 G   T
## Lines   total/split/joined/realigned/mismatch_removed/dup_removed/skipped:
10022/0/0/851/0/0/0
```

Sorting

BCFtools has a `sort` function but the usage does not indicate the sorting order. We can create a subsetted and shuffled VCF file and check the order after sorting.

```
bcftools view -h eg/Pfeiffer.vcf > eg/Pfeiffer_header.vcf
bcftools view eg/Pfeiffer.vcf | perl -nle 'BEGIN { srand(1984) } if (/^#/){ next };
print if rand(1) < 0.01' | shuf > eg/Pfeiffer_shuf.txt

cat eg/Pfeiffer_header.vcf eg/Pfeiffer_shuf.txt > eg/Pfeiffer_shuf.vcf

bcftools view -H eg/Pfeiffer_shuf.vcf | head

## [W::vcf_parse_info] INFO 'GENE' is not defined in the header, assuming Type=String
## [W::vcf_parse_info] INFO 'INHERITANCE' is not defined in the header, assuming
Type=String
## [W::vcf_parse_info] INFO 'MIM' is not defined in the header, assuming Type=String
## [W::vcf_parse_format_dict2] FORMAT 'DS' at 10:123256215 is not defined in the
header, assuming Type=String
## [W::vcf_parse_format_dict2] FORMAT 'GL' at 10:123256215 is not defined in the
header, assuming Type=String
## 11   102562700   rs2509010   C   T   434.44  PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=1.347;DB;DP=26;Dels=0;FS=1.757;HRun=1;HaplotypeScore=0;MQ
0=0;MQ=60;MQRankSum=-0.593;QD=16.71;ReadPosRankSum=0.216;set=variant2    GT:AD:DP:GQ:PL
0/1:9,17:26:99:464,0,265
## 4    3117168 rs1936032   C   G   460.62  PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=0.617;DB;DP=41;Dels=0;FS=0;HRun=0;HaplotypeScore=0;MQ0=0;
MQ=59.54;MQRankSum=-0.46;QD=11.23;ReadPosRankSum=1.09;set=variant2   GT:AD:DP:GQ:PL
0/1:23,18:41:99:491,0,640
## 8    48962552    rs118090687 G   A   427.7   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=0.032;DB;DP=23;Dels=0;FS=7.402;HRun=1;HaplotypeScore=0;MQ
0=0;MQ=60;MQRankSum=-1.194;QD=18.6;ReadPosRankSum=2.098;set=variant2 GT:AD:DP:GQ:PL
0/1:8,15:23:99:458,0,162
## 7    6963564 .   T   C   67.81   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=2.054;DP=9;Dels=0;FS=0;HRun=3;HaplotypeScore=0.9997;MQ0=0
;MQ=60;MQRankSum=0.248;QD=7.53;ReadPosRankSum=1.203;set=variant2 GT:AD:DP:GQ:PL
0/1:6,3:9:97.81:98,0,161
## 5    33649807    rs4504387   C   A   442.09  PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=-2.398;DB;DP=43;Dels=0;FS=1.259;HRun=1;HaplotypeScore=0;M
Q0=0;MQ=58.58;MQRankSum=-0.158;QD=10.28;ReadPosRankSum=-0.816;set=variant2
GT:AD:DP:GQ:PL  0/1:23,20:43:99:472,0,649
## 4    10117728    rs35782983  G   A   132.12  PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=-1.954;DB;DP=13;Dels=0;FS=5.434;HRun=0;HaplotypeScore=0;M
Q0=0;MQ=55.03;MQRankSum=0.66;QD=10.16;ReadPosRankSum=0.312;set=variant2  GT:AD:DP:GQ:PL
0/1:6,7:13:99:162,0,172
## 19   46032735    .   CAG C   1722.19 PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=3.711;DP=53;FS=7.534;HRun=0;HaplotypeScore=176.423;MQ0=0;
MQ=60.38;MQRankSum=0.755;QD=32.49;ReadPosRankSum=0.238;set=variant   GT:AD:DP:GQ:PL
0/1:13,40:53:99:1761,0,633
## 7    18767343    rs1178127   A   G   94.06   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=-0.876;DB;DP=9;Dels=0;FS=0;HRun=1;HaplotypeScore=0;MQ0=0;
MQ=60;MQRankSum=-1.159;QD=10.45;ReadPosRankSum=-1.159;set=variant2   GT:AD:DP:GQ:PL
0/1:4,5:9:99:124,0,102
## 16   28913787    rs7189927   T   C   53.72   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=-0.684;DB;DP=8;Dels=0;FS=0;HRun=0;HaplotypeScore=0;MQ0=0;
MQ=60;MQRankSum=1.221;QD=6.72;ReadPosRankSum=0.956;set=variant2  GT:AD:DP:GQ:PL
0/1:5,3:8:83.72:84,0,164
## 22   30220925    rs112626077 C   T   68.88   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=-0.727;DB;DP=4;Dels=0;FS=0;HRun=0;HaplotypeScore=0;MQ0=0;
MQ=60;MQRankSum=0.727;QD=17.22;ReadPosRankSum=-0.727;set=variant2    GT:AD:DP:GQ:PL
0/1:1,3:4:26.54:99,0,27
```

Sort.

```
bcftools sort eg/Pfeiffer_shuf.vcf > eg/Pfeiffer_sorted.vcf
```

```
## Writing to /tmp/bcftools.F4WlSD
## Merging 1 temporary files
## Done
## Cleaning
```

Check first 10 lines.

```
bcftools view -H eg/Pfeiffer_sorted.vcf | head
```

```
## 1    10706167    .   G   A   519.75  PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=1.016;DP=24;Dels=0;FS=2.58;HRun=0;HaplotypeScore=0.8667;M
Q0=0;MQ=60;MQRankSum=0.953;QD=21.66;ReadPosRankSum=-0.635;set=variant2   GT:AD:DP:GQ:PL
0/1:7,17:24:99:550,0,150
## 1    21798249    rs72872297  C   G   693.02  PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=-2.637;DB;DP=70;Dels=0;FS=0;HRun=0;HaplotypeScore=4.3216;
MQ0=0;MQ=57.77;MQRankSum=0.804;QD=9.9;ReadPosRankSum=-0.816;set=variant2 GT:AD:DP:GQ:PL
0/1:39,31:70:99:723,0,1093
## 1    22168216    rs7556412   C   T   132.97  PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=2.367;DB;DP=12;Dels=0;FS=3.282;HRun=1;HaplotypeScore=0;MQ
0=0;MQ=60;MQRankSum=0.831;QD=11.08;ReadPosRankSum=2.141;set=variant2 GT:AD:DP:GQ:PL
0/1:7,5:12:99:163,0,159
## 1    24383819    rs7540491   T   C   355.27  PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=3.897;DB;DP=24;Dels=0;FS=9.582;HRun=0;HaplotypeScore=0.96
65;MQ0=0;MQ=60;MQRankSum=-0.895;QD=14.8;ReadPosRankSum=-0.26;set=variant2
GT:AD:DP:GQ:PL  0/1:12,12:24:99:385,0,307
## 1    24859767    rs12061734  A   C   297.1   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=2.862;DB;DP=23;Dels=0;FS=3.834;HRun=0;HaplotypeScore=0;MQ
0=0;MQ=58.36;MQRankSum=-0.277;QD=12.92;ReadPosRankSum=0.277;set=variant2 GT:AD:DP:GQ:PL
0/1:11,12:23:99:327,0,246
## 1    25098335    rs77984282  CA  C   2179.77 PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=6.102;DB;DP=138;FS=2.162;HRun=2;HaplotypeScore=817.205;MQ
0=0;MQ=59.36;MQRankSum=0.514;QD=15.8;ReadPosRankSum=1.164;set=variant    GT:AD:DP:GQ:PL
0/1:70,68:138:99:2219,0,2674
## 1    34006598    rs12741094  C   G   69.33   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=-0.55;DB;DP=7;Dels=0;FS=0;HRun=0;HaplotypeScore=0;MQ0=0;M
Q=60;MQRankSum=-0.55;QD=9.9;ReadPosRankSum=-1.3;set=variant2 GT:AD:DP:GQ:PL
0/1:4,3:7:99:99,0,142
## 1    34189957    rs11346874  AG  A   210.16  PASS
AC=2;AF=1;AN=2;DB;DP=7;FS=0;HRun=4;HaplotypeScore=18.582;MQ0=0;MQ=60;QD=30.02;set=varia
nt   GT:AD:DP:GQ:PL  1/1:0,7:7:21.06:252,21,0
## 1    36641729    rs2296263   C   T   139.4   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=0.387;DB;DP=11;Dels=0;FS=0;HRun=1;HaplotypeScore=0;MQ0=0;
MQ=60;MQRankSum=1.186;QD=12.67;ReadPosRankSum=-0.466;set=variant2    GT:AD:DP:GQ:PL
0/1:6,5:11:99:169,0,141
## 1    45444038    rs11556200  G   A   766.85  PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=1.513;DB;DP=82;Dels=0;FS=0;HRun=0;HaplotypeScore=0;MQ0=0;
MQ=60;MQRankSum=-0.842;QD=9.35;ReadPosRankSum=-1.532;set=variant2    GT:AD:DP:GQ:PL
0/1:49,33:82:99:797,0,1220
```

Check chromosome order.

```
bcftools view -H eg/Pfeiffer_sorted.vcf | cut -f1 | uniq
```

```
## 1
## 10
## 11
## 12
## 13
## 14
## 15
## 16
## 17
## 18
## 19
## 2
## 20
## 21
## 22
## 3
## 4
## 5
## 6
## 7
## 8
## 9
## GL000220.1
## GL000225.1
## GL000231.1
## GL000241.1
## X
```

BCFtools sorts by lexical order (and there does not seem to be an option for another order).

If you use Linux, we can use the `sort` utility provided by your distribution instead of `bcftools sort`; we just need to separate and join the VCF header back. For this README, the following version of `sort` was used.

```
sort --version
```

```
## sort (GNU coreutils) 8.30
## Copyright (C) 2018 Free Software Foundation, Inc.
## License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
## This is free software: you are free to change and redistribute it.
## There is NO WARRANTY, to the extent permitted by law.
##
## Written by Mike Haertel and Paul Eggert.
```

Sort using natural sort.

```
sort -k1,1V -k2,2n eg/Pfeiffer_shuf.txt > eg/Pfeiffer_ns.txt
cat eg/Pfeiffer_header.vcf eg/Pfeiffer_ns.txt >  eg/Pfeiffer_ns.vcf
bcftools view -H eg/Pfeiffer_ns.vcf | cut -f1 | uniq
```

```
## 1
## 2
## 3
## 4
## 5
## 6
## 7
## 8
## 9
## 10
## 11
## 12
## 13
## 14
## 15
## 16
## 17
## 18
## 19
## 20
## 21
## 22
## GL000220.1
## GL000225.1
## GL000231.1
## GL000241.1
## X
```

Index a VCF file

An index is required for several tasks such as . Using either `bcftools index` or `tabix` seems to be fine, although they generate indexes with different suffixes.

Using `bcftools index` to generate a `csi` index.

```
bcftools index eg/aln.hc.vcf.gz
bcftools view -H -r 1000000:100-1000 eg/aln.hc.vcf.gz
rm eg/aln.hc.vcf.gz.csi
```

```
## 1000000  151 .   T   A   4111.06 .
AC=2;AF=1;AN=2;DP=100;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=25.36;SOR=9.065
GT:AD:DP:GQ:PL  1/1:0,92:92:99:4125,277,0
## 1000000  172 .   TA  T   3122.03 .
AC=2;AF=1;AN=2;DP=89;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=28.73;SOR=8.909
GT:AD:DP:GQ:PL  1/1:0,85:85:99:3136,256,0
## 1000000  336 .   A   G   4884.06 .
AC=2;AF=1;AN=2;DP=115;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=30.97;SOR=9.401
GT:AD:DP:GQ:PL  1/1:0,109:109:99:4898,328,0
## 1000000  415 .   C   A   4569.06 .
AC=2;AF=1;AN=2;DP=106;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=27.24;SOR=3.442
GT:AD:DP:GQ:PL  1/1:0,102:102:99:4583,307,0
## 1000000  733 .   G   T   9279.06 .
AC=2;AF=1;AN=2;DP=215;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=28.2;SOR=0.814
GT:AD:DP:GQ:PL  1/1:0,208:208:99:9293,626,0
```

Using `tabix` to generate a `tbi` index.

```
tabix eg/aln.hc.vcf.gz
bcftools view -H -r 1000000:100-1000 eg/aln.hc.vcf.gz
rm eg/aln.hc.vcf.gz.tbi
```

```
## 1000000  151 .   T   A   4111.06 .
AC=2;AF=1;AN=2;DP=100;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=25.36;SOR=9.065
GT:AD:DP:GQ:PL  1/1:0,92:92:99:4125,277,0
## 1000000  172 .   TA  T   3122.03 .
AC=2;AF=1;AN=2;DP=89;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=28.73;SOR=8.909
GT:AD:DP:GQ:PL  1/1:0,85:85:99:3136,256,0
## 1000000  336 .   A   G   4884.06 .
AC=2;AF=1;AN=2;DP=115;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=30.97;SOR=9.401
GT:AD:DP:GQ:PL  1/1:0,109:109:99:4898,328,0
## 1000000  415 .   C   A   4569.06 .
AC=2;AF=1;AN=2;DP=106;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=27.24;SOR=3.442
GT:AD:DP:GQ:PL  1/1:0,102:102:99:4583,307,0
## 1000000  733 .   G   T   9279.06 .
AC=2;AF=1;AN=2;DP=215;ExcessHet=0;FS=0;MLEAC=2;MLEAF=1;MQ=60;QD=28.2;SOR=0.814
GT:AD:DP:GQ:PL  1/1:0,208:208:99:9293,626,0
```

Rename sample names

Use `reheader` to rename samples in a VCF using a text file containing the new sample names (one per line). Note that the default output format for `reheader` is BCF (with no option for another format).

Old sample names.

```
bcftools query -l eg/1001genomes_snp-short-indel_only_ACGTN_5000.vcf.gz | head -5
```

```
## 88
## 108
## 139
## 159
## 265
```

Rename VCF file using new sample names in `eg/sample_name.txt`.

```
num_sample=$(bcftools query -l eg/1001genomes_snp-short-indel_only_ACGTN_5000.vcf.gz |
wc -l)
for (( i = 0 ; i < ${num_sample} ; i++ )); do echo ${i}; done > eg/sample_name.txt

bcftools reheader -s eg/sample_name.txt eg/1001genomes_snp-short-
indel_only_ACGTN_5000.vcf.gz | bcftools view -h - | tail -1
```

```
## #CHROM   POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  0   1   2   3   4   5   6
7   8   9   10  11  12  13  14  15  16  17  18  19  20  21  22  23  24  25  26  27  28
29  30  31  32  33  34  35  36  37  38  39  40  41  42  43  44  45  46  47  48  49  50
51  52  53  54  55  56  57  58  59  60  61  62  63  64  65  66  67  68  69  70  71  72
73  74  75  76  77  78  79  80  81  82  83  84  85  86  87  88  89  90  91  92  93  94
95  96  97  98  99  100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116
117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138
139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160
161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182
183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204
205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226
227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248
249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270
271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314
315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336
337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358
359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380
381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402
403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424
425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446
447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468
469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490
491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512
513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534
535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556
557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578
579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600
601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622
623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644
645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666
667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688
689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710
711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732
733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754
755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776
777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798
799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820
821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842
843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864
865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886
887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908
909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930
931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952
953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974
975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996
997 998 999 1000    1001    1002    1003    1004    1005    1006    1007    1008
1009    1010    1011    1012    1013    1014    1015    1016    1017    1018    1019
1020    1021    1022    1023    1024    1025    1026    1027    1028    1029    1030
1031    1032    1033    1034    1035    1036    1037    1038    1039    1040    1041
1042    1043    1044    1045    1046    1047    1048    1049    1050    1051    1052
1053    1054    1055    1056    1057    1058    1059    1060    1061    1062    1063
1064    1065    1066    1067    1068    1069    1070    1071    1072    1073    1074
1075    1076    1077    1078    1079    1080    1081    1082    1083    1084    1085
1086    1087    1088    1089    1090    1091    1092    1093    1094    1095    1096
1097    1098    1099    1100    1101    1102    1103    1104    1105    1106    1107
```

```
1108    1109    1110    1111    1112    1113    1114    1115    1116    1117    1118
1119    1120    1121    1122    1123    1124    1125    1126    1127    1128    1129
1130    1131    1132    1133    1134
```

Remove header info

The `reheader` command can also be used to modify the header information, such as removing specific info. For example if we want to remove `GATKCommandLine`, we would first create a new header file and use it to generate a new header.

`GATKCommandLine` exists in the header before running `reheader`.

```
bcftools view -h eg/aln.hc.vcf.gz | grep GATKCommandLine
```

```
## ##GATKCommandLine=<ID=HaplotypeCaller,CommandLine="HaplotypeCaller --output
aln.hc.vcf --input aln.bam --reference test_31.fa --use-posteriors-to-calculate-qual
false --dont-use-dragstr-priors false --use-new-qual-calculator true --annotate-with-
num-discovered-alleles false --heterozygosity 0.001 --indel-heterozygosity 1.25E-4 --
heterozygosity-stdev 0.01 --standard-min-confidence-threshold-for-calling 30.0 --max-
alternate-alleles 6 --max-genotype-count 1024 --sample-ploidy 2 --num-reference-
samples-if-no-call 0 --genotype-assignment-method USE_PLS_TO_ASSIGN --contamination-
fraction-to-filter 0.0 --output-mode EMIT_VARIANTS_ONLY --all-site-pls false --gvcf-gq-
bands 1 --gvcf-gq-bands 2 --gvcf-gq-bands 3 --gvcf-gq-bands 4 --gvcf-gq-bands 5 --gvcf-
gq-bands 6 --gvcf-gq-bands 7 --gvcf-gq-bands 8 --gvcf-gq-bands 9 --gvcf-gq-bands 10 --
gvcf-gq-bands 11 --gvcf-gq-bands 12 --gvcf-gq-bands 13 --gvcf-gq-bands 14 --gvcf-gq-
bands 15 --gvcf-gq-bands 16 --gvcf-gq-bands 17 --gvcf-gq-bands 18 --gvcf-gq-bands 19 --
gvcf-gq-bands 20 --gvcf-gq-bands 21 --gvcf-gq-bands 22 --gvcf-gq-bands 23 --gvcf-gq-
bands 24 --gvcf-gq-bands 25 --gvcf-gq-bands 26 --gvcf-gq-bands 27 --gvcf-gq-bands 28 --
gvcf-gq-bands 29 --gvcf-gq-bands 30 --gvcf-gq-bands 31 --gvcf-gq-bands 32 --gvcf-gq-
bands 33 --gvcf-gq-bands 34 --gvcf-gq-bands 35 --gvcf-gq-bands 36 --gvcf-gq-bands 37 --
gvcf-gq-bands 38 --gvcf-gq-bands 39 --gvcf-gq-bands 40 --gvcf-gq-bands 41 --gvcf-gq-
bands 42 --gvcf-gq-bands 43 --gvcf-gq-bands 44 --gvcf-gq-bands 45 --gvcf-gq-bands 46 --
gvcf-gq-bands 47 --gvcf-gq-bands 48 --gvcf-gq-bands 49 --gvcf-gq-bands 50 --gvcf-gq-
bands 51 --gvcf-gq-bands 52 --gvcf-gq-bands 53 --gvcf-gq-bands 54 --gvcf-gq-bands 55 --
gvcf-gq-bands 56 --gvcf-gq-bands 57 --gvcf-gq-bands 58 --gvcf-gq-bands 59 --gvcf-gq-
bands 60 --gvcf-gq-bands 70 --gvcf-gq-bands 80 --gvcf-gq-bands 90 --gvcf-gq-bands 99 --
floor-blocks false --indel-size-to-eliminate-in-ref-model 10 --disable-optimizations
false --dragen-mode false --apply-bqd false --apply-frd false --disable-spanning-event-
genotyping false --transform-dragen-mapping-quality false --mapping-quality-threshold-
for-genotyping 20 --max-effective-depth-adjustment-for-frd 0 --just-determine-active-
regions false --dont-genotype false --do-not-run-physical-phasing false --do-not-
correct-overlapping-quality false --use-filtered-reads-for-annotations false --
adaptive-pruning false --do-not-recover-dangling-branches false --recover-dangling-
heads false --kmer-size 10 --kmer-size 25 --dont-increase-kmer-sizes-for-cycles false -
-allow-non-unique-kmers-in-ref false --num-pruning-samples 1 --min-dangling-branch-
length 4 --recover-all-dangling-branches false --max-num-haplotypes-in-population 128 -
-min-pruning 2 --adaptive-pruning-initial-error-rate 0.001 --pruning-lod-threshold
2.302585092994046 --pruning-seeding-lod-threshold 9.210340371976184 --max-unpruned-
variants 100 --linked-de-bruijn-graph false --disable-artificial-haplotype-recovery
false --enable-legacy-graph-cycle-detection false --debug-assembly false --debug-graph-
transformations false --capture-assembly-failure-bam false --num-matching-bases-in-
dangling-end-to-recover -1 --error-correction-log-odds -Infinity --error-correct-reads
false --kmer-length-for-read-error-correction 25 --min-observations-for-kmer-to-be-
solid 20 --base-quality-score-threshold 18 --dragstr-het-hom-ratio 2 --dont-use-
dragstr-pair-hmm-scores false --pair-hmm-gap-continuation-penalty 10 --expected-
mismatch-rate-for-read-disqualification 0.02 --pair-hmm-implementation
FASTEST_AVAILABLE --pcr-indel-model CONSERVATIVE --phred-scaled-global-read-mismapping-
rate 45 --disable-symmetric-hmm-normalizing false --disable-cap-base-qualities-to-map-
quality false --enable-dynamic-read-disqualification-for-genotyping false --dynamic-
read-disqualification-threshold 1.0 --native-pair-hmm-threads 4 --native-pair-hmm-use-
double-precision false --bam-writer-type CALLED_HAPLOTYPES --dont-use-soft-clipped-
bases false --min-base-quality-score 10 --smith-waterman JAVA --emit-ref-confidence
NONE --max-mnp-distance 0 --force-call-filtered-alleles false --soft-clip-low-quality-
ends false --allele-informative-reads-overlap-margin 2 --smith-waterman-dangling-end-
match-value 25 --smith-waterman-dangling-end-mismatch-penalty -50 --smith-waterman-
dangling-end-gap-open-penalty -110 --smith-waterman-dangling-end-gap-extend-penalty -6
--smith-waterman-haplotype-to-reference-match-value 200 --smith-waterman-haplotype-to-
reference-mismatch-penalty -150 --smith-waterman-haplotype-to-reference-gap-open-
penalty -260 --smith-waterman-haplotype-to-reference-gap-extend-penalty -11 --smith-
waterman-read-to-haplotype-match-value 10 --smith-waterman-read-to-haplotype-mismatch-
```

```
penalty -15 --smith-waterman-read-to-haplotype-gap-open-penalty -30 --smith-waterman-
read-to-haplotype-gap-extend-penalty -5 --min-assembly-region-size 50 --max-assembly-
region-size 300 --active-probability-threshold 0.002 --max-prob-propagation-distance 50
--force-active false --assembly-region-padding 100 --padding-around-indels 75 --
padding-around-snps 20 --padding-around-strs 75 --max-extension-into-assembly-region-
padding-legacy 25 --max-reads-per-alignment-start 50 --enable-legacy-assembly-region-
trimming false --interval-set-rule UNION --interval-padding 0 --interval-exclusion-
padding 0 --interval-merging-rule ALL --read-validation-stringency SILENT --seconds-
between-progress-updates 10.0 --disable-sequence-dictionary-validation false --create-
output-bam-index true --create-output-bam-md5 false --create-output-variant-index true
--create-output-variant-md5 false --max-variants-per-shard 0 --lenient false --add-
output-sam-program-record true --add-output-vcf-command-line true --cloud-prefetch-
buffer 40 --cloud-index-prefetch-buffer -1 --disable-bam-index-caching false --sites-
only-vcf-output false --help false --version false --showHidden false --verbosity INFO
--QUIET false --use-jdk-deflater false --use-jdk-inflater false --gcs-max-retries 20 --
gcs-project-for-requester-pays  --disable-tool-default-read-filters false --minimum-
mapping-quality 20 --disable-tool-default-annotations false --enable-all-annotations
false --allow-old-rms-mapping-quality-annotation-data
false",Version="4.2.4.1",Date="March 7, 2022 at 7:25:57 AM UTC">
```

`GATKCommandLine` has been removed with `reheader`.

```
bcftools view -h eg/aln.hc.vcf.gz | grep -v GATKCommandLine > eg/new_header.txt
bcftools reheader -h eg/new_header.txt eg/aln.hc.vcf.gz | bcftools view -h -
```

```
## ##fileformat=VCFv4.2
## ##FILTER=<ID=PASS,Description="All filters passed">
## ##FILTER=<ID=LowQual,Description="Low quality">
## ##FORMAT=<ID=AD,Number=R,Type=Integer,Description="Allelic depths for the ref and
alt alleles in the order listed">
## ##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth (reads
with MQ=255 or with bad mates are filtered)">
## ##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
## ##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
## ##FORMAT=<ID=PL,Number=G,Type=Integer,Description="Normalized, Phred-scaled
likelihoods for genotypes as defined in the VCF specification">
## ##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes, for each
ALT allele, in the same order as listed">
## ##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency, for each ALT
allele, in the same order as listed">
## ##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called
genotypes">
## ##INFO=<ID=BaseQRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon rank
sum test of Alt Vs. Ref base qualities">
## ##INFO=<ID=DP,Number=1,Type=Integer,Description="Approximate read depth; some reads
may have been filtered">
## ##INFO=<ID=ExcessHet,Number=1,Type=Float,Description="Phred-scaled p-value for exact
test of excess heterozygosity">
## ##INFO=<ID=FS,Number=1,Type=Float,Description="Phred-scaled p-value using Fisher's
exact test to detect strand bias">
## ##INFO=<ID=InbreedingCoeff,Number=1,Type=Float,Description="Inbreeding coefficient
as estimated from the genotype likelihoods per-sample when compared against the Hardy-
Weinberg expectation">
## ##INFO=<ID=MLEAC,Number=A,Type=Integer,Description="Maximum likelihood expectation
(MLE) for the allele counts (not necessarily the same as the AC), for each ALT allele,
in the same order as listed">
## ##INFO=<ID=MLEAF,Number=A,Type=Float,Description="Maximum likelihood expectation
(MLE) for the allele frequency (not necessarily the same as the AF), for each ALT
allele, in the same order as listed">
## ##INFO=<ID=MQ,Number=1,Type=Float,Description="RMS Mapping Quality">
## ##INFO=<ID=MQRankSum,Number=1,Type=Float,Description="Z-score From Wilcoxon rank sum
test of Alt vs. Ref read mapping qualities">
## ##INFO=<ID=QD,Number=1,Type=Float,Description="Variant Confidence/Quality by Depth">
## ##INFO=<ID=ReadPosRankSum,Number=1,Type=Float,Description="Z-score from Wilcoxon
rank sum test of Alt vs. Ref read position bias">
## ##INFO=<ID=SOR,Number=1,Type=Float,Description="Symmetric Odds Ratio of 2x2
contingency table to detect strand bias">
## ##contig=<ID=1000000,length=1000000>
## ##source=HaplotypeCaller
## ##bcftools_viewVersion=1.22+htslib-1.22
## ##bcftools_viewCommand=view -h eg/aln.hc.vcf.gz; Date=Mon Aug  4 00:42:43 2025
## ##bcftools_viewCommand=view -h -; Date=Mon Aug  4 00:42:43 2025
## #CHROM   POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  test
```

Concatenate VCF files

Concatenating is similar to joining VCF files in a row-wise manner. This only works for VCF files with the same samples that are also organised in the same order.

If we concatenate `eg/aln.bt.vcf.gz` and `eg/aln.hc.vcf.gz` we would expect that the total number of variants is the sum of both.

```
expr $(bcftools view -H eg/aln.bt.vcf.gz | wc -l) + $(bcftools view -H eg/aln.hc.vcf.gz
| wc -l)
```

```
## 19997
```

Perform the concatenation with default settings and count the number of variants.

```
bcftools concat eg/aln.bt.vcf.gz eg/aln.hc.vcf.gz  | bcftools view -H - | wc -l
```

```
## Checking the headers and starting positions of 2 files
## [W::bcf_hdr_merge] Trying to combine "MQ" tag definitions of different types
## Concatenating eg/aln.bt.vcf.gz   0.015467 seconds
## Concatenating eg/aln.hc.vcf.gz   0.017434 seconds
## 19997
```

Removing duplicates requires indexed VCF files; the `-f` parameter is used with `bcftools index` to overwrite an index if it exists.

```
bcftools index -f eg/aln.bt.vcf.gz
bcftools index -f eg/aln.hc.vcf.gz
```

To remove duplicates (`-D`) we need to allow overlaps (`-a`).

```
bcftools concat -a -D eg/aln.bt.vcf.gz eg/aln.hc.vcf.gz  | bcftools view -H - | wc -l
```

```
## Checking the headers and starting positions of 2 files
## [W::bcf_hdr_merge] Trying to combine "MQ" tag definitions of different types
## 10890
```

Merging VCF files

Use `bcftools merge` to merge VCF files. In this workflow, `bcftools merge` was used to create a multi-sample VCF file from individual VCF files.

```
bcftools merge -o PRJNA784038_illumina.vcf -O v SRR*.vcf.gz
```

Decomposing and normalising variants

Decomposing can refer to the splitting of multi-allelic variants; we can use `bcftools norm -m` for this.

```
bcftools view -H eg/PRJNA784038_illumina.vcf.gz | head -2 | cut -f1-5
```

```
## NC_045512.2  4   .   A   T
## NC_045512.2  16  .   C   A,G
```

Splitting.

```
bcftools norm -m- eg/PRJNA784038_illumina.vcf.gz | grep -v "^#" | head -3 | cut -f1-5
```

```
## NC_045512.2  4   .   A   T
## NC_045512.2  16  .   C   A
## NC_045512.2  16  .   C   G
```

Decomposing can also refer to converting MNVs into consecutive SNVs; this is achieved with `bcftools norm -a`.

```
bcftools view -H eg/ex.vcf  | tail -1
```

```
## 1    25563113    .   CC  GG  .   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=2.621;DB;DP=20;FS=0;HRun=0;HaplotypeScore=101.749;MQ0=0;M
Q=55.8;MQRankSum=-1.91;QD=11.05;ReadPosRankSum=0.4;set=variant   GT:AD:DP:GQ:PL
0/1:14,6:20:99:260,0,486    0/0:14,6:20:99:260,0,486    1/1:14,6:20:99:260,0,486
```

Convert MNV into SNVs.

```
bcftools norm -a eg/ex.vcf | tail -2
```

```
## Lines   total/split/joined/realigned/mismatch_removed/dup_removed/skipped:
5/0/0/0/0/0/0
## 1    25563113    .   C   G   .   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=2.621;DB;DP=20;FS=0;HRun=0;HaplotypeScore=101.749;MQ0=0;M
Q=55.8;MQRankSum=-1.91;QD=11.05;ReadPosRankSum=0.4;set=variant   GT:AD:DP:GQ:PL
0/1:14,6:20:99:260,0,486    0/0:14,6:20:99:260,0,486    1/1:14,6:20:99:260,0,486
## 1    25563114    .   C   G   .   PASS
AC=1;AF=0.5;AN=2;BaseQRankSum=2.621;DB;DP=20;FS=0;HRun=0;HaplotypeScore=101.749;MQ0=0;M
Q=55.8;MQRankSum=-1.91;QD=11.05;ReadPosRankSum=0.4;set=variant   GT:AD:DP:GQ:PL
0/1:14,6:20:99:260,0,486    0/0:14,6:20:99:260,0,486    1/1:14,6:20:99:260,0,486
```

Finally, there is also left-aligning, which will be clear by viewing an example.

```
bcftools view -H eg/aln.bt.vcf.gz | head -2
```

```
## 1000000  151 .   T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
## 1000000  172 .   TAA TA  142.185 .
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60 GT:PL   1/1:169,117,0
```

Left-align.

```
bcftools norm -f eg/test_31.fa eg/aln.bt.vcf.gz | grep -v "^#" | head -2
```

```
## 1000000  151 .   T   A   225.417 .
DP=92;VDB=0.696932;SGB=-0.693147;FS=0;MQ0F=0;AC=2;AN=2;DP4=0,0,90,0;MQ=60   GT:PL
1/1:255,255,0
## 1000000  172 .   TA  T   142.185 .
INDEL;IDV=75;IMF=0.914634;DP=82;VDB=0.712699;SGB=-0.693147;RPBZ=-4.35727;MQBZ=0;SCBZ=0;
FS=0;MQ0F=0;AC=2;AN=2;DP4=7,0,75,0;MQ=60 GT:PL   1/1:169,117,0
```

Decomposing and normalising variants are very important steps when comparing VCF files.

Comparing VCF files

BCFtools has an intersect tool (`bcftools isec`) that is useful for comparing VCF files; the tool requires an index.

```
tabix -f eg/aln.bt.vcf.gz
tabix -f eg/aln.hc.vcf.gz

bcftools isec eg/aln.bt.vcf.gz eg/aln.hc.vcf.gz -p comp
```

Four VCF files are produced and `README.txt` explains the contents of each VCF file.

Records private to `eg/aln.bt.vcf.gz`.

```
bcftools view -H comp/0000.vcf | wc -l

## 915
```

Records private to `eg/aln.hc.vcf.gz`.

```
bcftools view -H comp/0001.vcf | wc -l

## 868
```

Number of overlap.

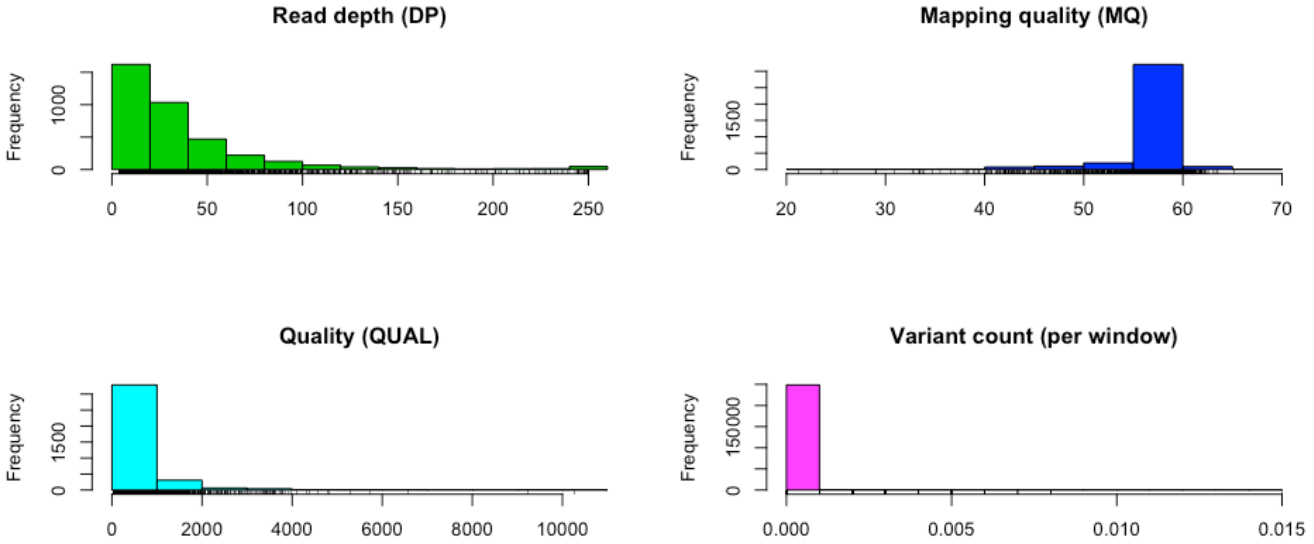
```
bcftools view -H comp/0002.vcf | wc -l

## 9107
```

SnpSift is also useful for comparing VCF files for checking their concordance.

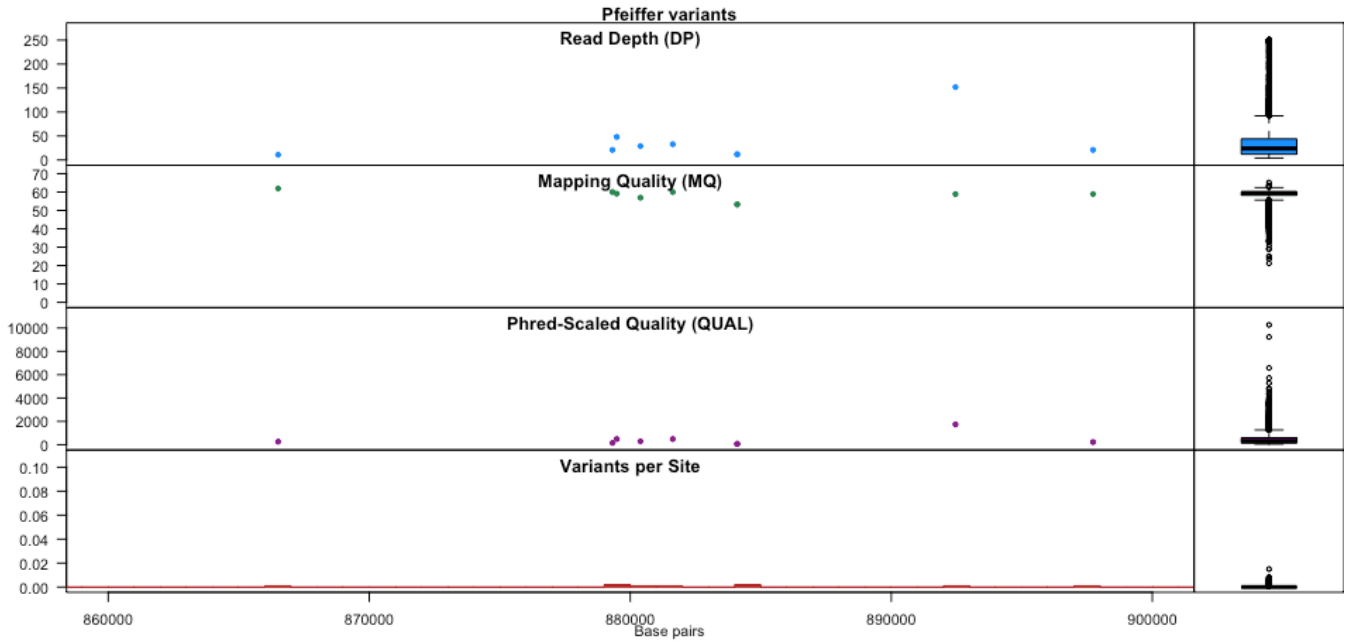
Visualisation

The vcfR package produces some nice plots.

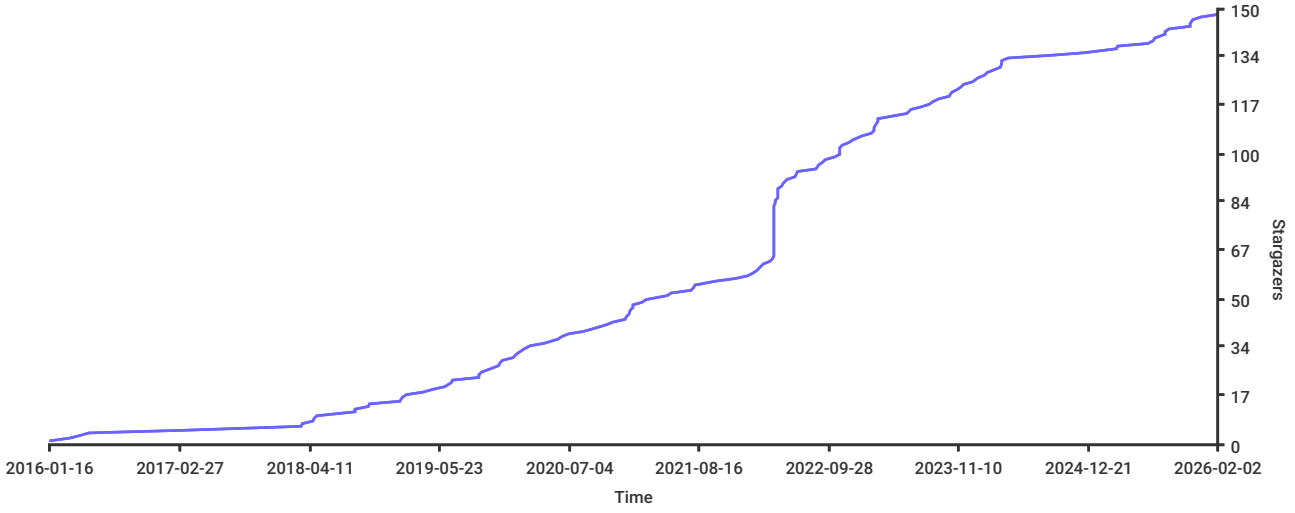
```
install.packages("vcfR")
library(vcfR)

my_vcf <- read.vcfR("~/github/learning_vcf_file/eg/Pfeiffer.vcf", verbose = FALSE)
chrom <- create.chromR(name="Pfeiffer variants", vcf=my_vcf)
chrom <- proc.chromR(chrom, verbose=TRUE)

plot(chrom)
```




```
chromoqc(chrom, xlim=c(860000, 900000))
```



Useful links

- A very useful thread on SEQanswers on learning about the VCF format: http://seqanswers.com/forums/showthread.php?t=9345
- Useful tutorial on VCFs files from the 1000 Genomes Project Page: http://www.1000genomes.org/node/101
- The author of ANNOVAR writes about VCF files: http://annovar.openbioinformatics.org/en/latest/articles/VCF/
- Encoding Structural Variants in VCF version 4.0
- BCFtools cheat sheet

Stargazers over time